Annexes

I . tableau des valeurs de Dcritique. pour le test K-S

TABLE FOR KOLMOGOROV-SMIRNOV ONE-SAMPLE TEST:
CRITICAL VALUES FOR D_{max} STATISTIC

n	$\alpha = 0.20$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.02$	$\alpha = 0.01$
1	0,900	0,950	0,975	0,99	0,995
2	0,684	0,776	0,842	0,9	0,929
3	0,565	0,636	0,708	0,785	0,829
4	0,493	0,656	0,624	0,689	0,734
5	0,447	0,509	0,563	0,627	0,669
6	0,410	0,468	0,519	0,577	0,617
7	0,381	0,436	0,483	0,538	0,576
8	0,358	0,41	0,454	0,507	0,542
9	0,339	0,387	0,43	0,48	0,513
10	0,323	0,369	0,409	0,457	0,489
11	0,308	0,352	0,391	0,437	0,468
12	0,296	0,338	0,375	0,419	0,449
13	0,285	0,325	0,361	0,404	0,432
14	0,275	0,314	0,349	0,39	0,418
15	0,266	0,304	0,338	0,377	0,404
16	0,258	0,295	0,327	0,366	0,392
17	0,250	0,286	0,318	0,355	0,381
18	0,244	0,279	0,309	0,346	0,371
19	0,237	0,271	0,301	0,337	0,361
20	0,232	0,265	0,294	0,329	0,352
21	0,226	0,259	0,287	0,321	0,344
22	0,221	0,253	0,281	0,314	0,337
23	0,216	0,247	0,275	0,307	0,33
24	0,212	0,242	0,269	0,301	0,323
25	0,208	0,238	0,264	0,295	0,317
26	0,204	0,233	0,259	0,29	0,311
27	0,200	0,229	0,254	0,284	0,305
28	0,197	0,225	0,25	0,279	0,300
29	0,193	0,221	0,246	0,275	0,295
30	0,19	0,218	0,242	0,270	0,290
31	0,187	0,214	0,238	0,266	0,285
32	0,184	0,211	0,234	0,262	0,281
33	0,182	0,208	0,231	0,258	0,277
34	0,179	0,205	0,227	0,254	0,273
35	0,177	0,202	0,224	0,251	0,269
36	0,174	0,199	0,221	0,247	0,265
37	0,172	0,196	0,218	0,244	0,262
38	0,170	0,194	0,215	0,241	0,258
39	0,168	0,191	0,213	0,238	0,255
40	0,165	0,189	0,210	0,235	0,252
> 40	$1.07/\sqrt{n}$	$1.22/\sqrt{n}$	$1.36/\sqrt{n}$	$1.52/\sqrt{n}$	$1.63/\sqrt{n}$

Compiled and edited by Josemari Sarasola for

Available at: https://gizapedia.hirusta.io/pdf-printer-friendly-statistical-table-for-the-one-sample-kolmogorov-s

Figure 1 : Valeurs Dcritique (test de Kolmogorov-Smirnov)

II. Données des pluies de 5, 10, 20, 30 et 60 minutes

STATION PLUVIOMETRIQUE DE BINZA :

Altitude : 445 m
Indicatif : 64220

INTENSITE MAXIMALE DES PLUIES en mm.

1, QUANTITE D'EAU ENREGISTREES EN 5 minutes.

STATION DE KIN-BINZA

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D	MAX. ANN
1969	3.3	6.0	7.4	17.5	2.3	0.0	0.0	0.0	1.9	5.0	6.3	4.0	17.5
1970	6.0	10.0	6.1	9.0	5.9	1.0	0.0	0.0	0.0	6.0	8.9	7.0	10.0
1971	4.9	3.1	9.0	10.0	8.0	0.0	0.0	0.0	4.1	5.0	7.5	5.4	10.0
1972	10.0	8.0	11.3	14.5	1.0	0.0	0.0	0.0	2.8	6.2	6.2	4.0	14.5
1973	9.5	9.0	3.5	7.3	3.5	0.0	0.0	0.0	3.5	6.0	10.0	8.0	10.0
1974	3.4	2.7	3.0	6.8	17.2	4.3	0.0	0.0	2.6	6.0	5.8	5.5	17.2
1975	5.0	9.7	19.9	8.8	3.5	0.0	0.2	0.1	1.5	5.3	7.5	4.2	19.9
1976	3.5	10.0	10.0	8.0	5.5	0.0	2.0	0.0	3.6	8.0	9.5	6.5	10.0
1977	10.2	9.2	9.4	10.5	18.6	9.4	0.0	0.0	4.7	9.9	11.0	9.3	18.6
1978	9.9	5.7	4.5	11.3	2.7	0.0	0.0	0.0	8.5	6.5	10.0	9.8	11.3
1979	9.8	7.4	4.4	10.0	6.7	0.0	0.0	0.0	2.0	6.7	10.6	10.6	10.6
1980	10.0	10.4	10.4	10.4	4.5	9.9	0.0	5.0	3.8	6.5	17.8	11.5	17.8
1981	10.1	11.0	14.0	15.5	2.5	0.1	0.0	1.9	3.6	8.0	10.0	7.0	15.5
1982	8.0	10.0	10.0	12.9	11.5	0.0	0.0	0.0	4.2	5.5	10.5	6.0	12.9
1983	7.0	16.0	10.0	10.0	8.0	0.0	2.0	0.0	0.0	8.8	10.0	8.0	16.0
1984	2.5	10.0	14.2	8.0	5.5	0.0	3.5	5.0	3.4	5.8	9.5	6.5	14.2
1985	8.8	2.0	11.5	10.0	9.5	0.0	4.4	0.1	8.5	6.6	9.8	10.0	11.5
1986	8.0	2.7	3.0	7.5	9.5	4.3	0.0	0.0	6.0	7.0	9.0	10.2	10.2
1987	6.0	10.0	12.0	9.9	4.5	0.0	0.8	0.1	5.3	6.7	11.4	5.0	12.0
1988	8.3	10.0	17.0	10.0	2.1	0.4	3.1	3.4		12.0	10.0	12.5	17.0
1989	10.0	10.0	8.0	11.0	6.4	0.4	0.0	0.7	2.0	5.7	9.0	14.5	14.5
1990	7.0	10.0	10.0	6.0	13.6	0.0	0.0	0.0	2.0	9.8	11.0	9.2	13.6
1991	8.4	6.0	7.0	6.8	7.0	0.3	0.0	3.5	2.2	4.7	11.8	12.8	12.8
1992	8.3	9.7	6.2	6.1	10.1	0.0	0.0	0.0	9.0	9.2	8.0	2.6	10.1
1983	5.0	1.5	12.7	10.0	3.0	0.0	0.0	0.0	1.7	6.0	10.0	3.2	12.7
1994	**	0.5	3.0	6.0	3.4	0.0	0.0			1.2	8.9	9.0	9.0
1995	5.0	9.8	9.8	2.3	5.8	0.0	0.0	6.0	10.0	5.0	10.0	14.0	14.0
1996	9.0	3.0	7.1	5.0	10.8	0.1	0.0	0.0	3.5	5.5	6.0	5.0	10.8
1997	8.0	4.0	5.0	5.5	7.0	0.3	0.0	0.8	0.2	12.0	9.9	8.0	12.0
1998	5.0	8.2	19.3	9.8	4.0	0.0	0.0	0.0	7.0	5.0	4.0	5.0	19.3
1999	9.9	8.2	9.8	4.6	6.0	1.8	0.2	0.0		12.5		12.3	12.5
2000	19.8	5.8	5.2	6.9	6.0	0.0	0.0	0.0	6.7	10.0	10.0	9.9	19.8

2, QUANTITE D'EAU ENREGISTREE EN 10 minutes.

STATION DE KIN-BINZA

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D	MAX. ANN
1969	20.0	12.1	18.5	9.7	7.4	0.0	0.1	0.0	5.8	9.0	13.5	11.0	20.0
1970	7.7	18.9	9.8	14.0	8.9	0.0	2.0	0.0	0.0	12.5	17.0	9.0	18.9
1971	22.3	6.3	12.5	16.8	10.5	0.0	0.0	0.0	2.2	9.5	15.5	12.3	22.3
1972	12.0	10.5	14.3	16.0	30.2	0.0	0.0	0.0	3.6	11.5	22.0	6.2	30.2
1973	13.5	10.4	5.3	30.7	6.5	0.0	0.0	0.0	5.8	7.8	15.0	10.0	30.7
1974	5.3	4.7	5.3	8.7	28.9	4.3	0.0	0.0	4.7	8.4	8.6	11.7	28.9
1975	9.6	11.5	18.4	15.0	4.3	0.0	1.5	0.2	2.5	8.2	14.4	7.7	18.4
1976	18.2	20.0	13.0	11.6	7.5	0.0	3.0	0.0	5.0	12.1	14.7	11.5	20.0
1977	15.2	14.5	17.4	18.4	27.9	16.5	0.0	0.0	8.2	12.8	19.9	13.3	27.9
1978	19.8	10.7	7.4	22.2	4.3	0.0	0.0	0.0	9.3	10.6	20.0	15.8	22.2
1979	18.4	12.8	6.6	18.4	11.4	0.0	0.0	0.0	3.9	10.3	21.2	21.2	21.2
1980	15.5	14.0	20.8	15.5	6.8	14.9	0.0	5.8	6.0	8.7	19.3	17.5	20.8
1981	15.0	15.5	26.0	30.0	3.5	0.1	0.0	2.1	5.0	12.1	25.0	7.0	30.0
1982	9.3	14.5	17.0	18.9	20.0	0.0	0.0	0.0	5.7	10.0	17.6	12.9	20.0
1983	10.0	25.0	13.0	16.0	11.0	0.0	3.0	0.0	0.0	15.0	18.0	10.0	25.0
1984	7.2	20.0	21.2	11.6	7.5	0.0	6.4	7.0	7.2	9.3	14.7	11.5	21.2
1985	12.0	3.1	20.7	20.0	16.5	0.0	8.0	0.1	12.0	8.6	21.7	18.0	21.7
1986	10.7	4.7	5.3	9.9	17.3	6.0	0.0	0.0	9.0	14.5	13.0	18.1	18.1
1987	9.6	15.7	17.6	15.9	7.2	0.0	1.5	0.2	7.5	12.2	17.2	8.3	17.6
1988	13.3	21.4	34.0	13.5	3.4	0.6	5.5	5.0		16.1	17.0	19.5	34.0
1989	13.5	14.5	11.5	21.5	10.0	0.6	0.0	1.0	3.5	9.7	16.0	22.0	22.0
1990	10.0	20.0	18.0	8.8	25.6	0.0	0.0	0.1	4.0	15.0	18.3	11.5	25.6
1991	13.0	9.0	13.0	8.3	12.0	0.4	0.0	4.1	4.0	7.7	18.9	18.9	18.9
1992	9.7	17.4	9.8	11.1	16.5	0.0	0.0	0.0	14.8	10.0	10.0	7.9	17.4
1993	6.9	3.5	19.8	20.0	8.0	0.0	0.0	0.0	2.0	10.0	10.4	6.8	20.0
1994		0.2	6.0	11.8	6.2	0.0	0.0			3.0	17.8	9.8	17.8
1995	9.9	13.9	19.8	5.0	13.0	0.0	0.0	10.0	15.0	9.8	20.0	23.8	23.8
1996	15.0	4.3	18.8	9.9	12.8	0.1	0.0	0.0	3.8	9.0	10.0	10.0	18.8
1997	11.6	10.1	10.0	24.0	9.8	0.8	0.0	1.0	1.0	19.2	11.3	11.4	24.0
1998	14.2	13.7	24.3	13.7	9.8	0.0	0.0	0.0	10.0	7.7	11.6	10.0	24.3
1999	14.4	18.6	19.8	7.5	13.3	4.5	0.4	0.0		27.3		20.3	27.3
2000	29.7	13.2	10.0	12.1	10.2	0.0	0.0	0.0	11.8	11.7	20.0	20.6	29.7
2001	3.7	28.5	20.4		27.0								28.5
2002													0.0

3, QUANTITE D'EAU ENREGISTREES EN 20 minutes.

STATION DE KIN-BINZA

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D	MAX. ANN
1969	40.3	33.1	20.0	13.7	9.7	0.0	0.0	0.0	9.7	12.8	11.8	12.5	40.3
1970	15.0	26.0	13.6	35.6	11.9	0.0	2.0	0.0	0.0	21.0	26.0	17.5	35.6
1971	12.5	6.5	42.3	32.9	22.0	0.0	0.0	0.0	2.6	18.5	23.9	19.2	42.3
1972	13.0	18.0	37.5	25.5	50.1	0.0	0.0	0.0	4.2	13.4	25.2	9.7	50.1
1973	14.6	17.8	8.2	52.0	6.1	0.0	0.0	0.0	5.8	10.3	29.4	21.0	52.0
1974	13.3	6.7	9.5	13.7	39.2	4.3	0.0	0.0	6.9	8.7	10.5	13.0	39.2
1975	33.0	25.7	33.9	22.7	3.5	0.0	2.5	1.1	4.4	12.1	17.6	10.8	33.9
1976	30.5	35.0	20.7	14.6	9.8	0.0	3.4	0.0	6.7	16.6	19.4	15.0	35.0
1977	20.4	26.4	26.7	25.3	37.3	18.7	0.0	0.0	10.9	17.2	24.5	17.3	37.3
1978	29.6	15.2	8.9	32.0	6.7	0.0	0.0	0.0	18.2	18.4	30.0	29.6	32.0
1979	28.2	18.7	10.2	19.8	17.8	0.0	0.0	0.0	4.6	15.8	34.2	31.8	34.2
1980	25.5	25.6	25.5	30.0	7.2	17.9	0.0	5.9	9.2	16.2	37.4	23.2	37.4
1981	22.5	24.5	49.5	45.6	4.1	0.2	0.0	2.8	6.7	16.6	37.7	13.1	49.5
1982	17.7	19.5	25.8	33.8	25.2	0.0	0.0	0.0	8.2	19.5	29.5	21.5	33.8
1983	18.0	28.0	20.7	24.5	12.2	0.0	3.4	0.0	0.0	21.5	28.0	20.0	28.0
1984	11.9	35.0	27.5	14.6	9.8	0.0	9.3	11.1	11.2	13.2	19.4	15.0	35.0
1985	22.6	4.8	31.7	26.5	24.7	0.0	9.4	0.2	14.9	17.5	26.5	30.1	31.7
1986	20.9	8.2	10.7	23.6	37.4	12.1	0.0	0.0	19.0	34.9	30.0	31.6	37.4
1987	16.0	28.7	27.5	24.0	7.5	0.0	2.5	0.6	11.5	14.0	31.3	16.3	31.3
1988	22.3	30.7	15.0	25.6	3.5	0.8	7.0	7.0		28.6	30.0	32.0	32.0
1989	23.1	25.5	15.3	31.7	11.3	0.8	0.0	1.4	5.0	19.2	21.5	29.7	31.7
1990	17.5	38.0	22.6	13.2	35.0	0.0	0.0	0.0	6.7	20.6	27.0	18.2	38.0
1991	22.0	18.8	21.4	18.0	24.0	0.0	0.0	7.2	6.0	14.4	25.5	25.4	25.5
1992	19.4	20.0	17.7	14.9	23.5	0.0	0.0	0.0	22.3	16.9	17.5	9.9	23.5
1993	10.5	5.4	27.0	27.1	14.1	0.0	0.0	0.0	4.0	20.0	28.5	9.2	28.5
1994		1.4	3.0	20.0	13.2	0.0	0.0			6.2	29.1	19.6	29.1
1995	20.8	32.0	35.4	9.3	21.7	0.0	0.0	23.5	27.0	13.9	39.0	34.0	39.0
1996	23.8	7.1	36.6	29.5	23.3	0.2	0.0	0.0	4.5	11.5	14.7	23.0	36.6
1997	20.6	18.1	26.0	28.0	17.8	1.2	0.0	2.3	1.1	26.0	13.9	18.9	28.0
1998	20.3	25.9	40.7	23.1	16.8	0.0	0.0	0.0	22.2	17.8	19.8	24.9	40.7
1999	21.4	24.6	39.6	11.3	14.7	6.8	0.6	0.0		32.4		36.2	39.6
2000	43.8	23.2	18.9	27.2	13.5	0.0	0.0	0.0	23.0	20.0	37.0	31.7	43.8
2001	8.0	36.5	24.8		29.4								36.5
2002													0.0
MAXIMA	29.6	38.0	49.5	45.6	37.4	18.7	9.4	23.5	27.0	34.9	39.0	34.0	49.5

4. QUANTITE D'EAU ENREGISTREE EN 30 minutes.

STATION DE KIN-BINZA

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D	MAX. ANN
1969	54.8	17.6	27.5	14.5	8.8	0.0	0.0	0.0	5.0	14.0	20.7	13.9	54.8
1970	16.5	35.0	18.5	48.9	10.8	0.0	3.2	0.0	0.0	25.5	30.4	18.5	48.9
1971	56.1	15.7	24.7	42.0	25.9	0.0	0.0	0.0	2.5	29.5	31.2	23.1	56.1
1972	22.3	25.0	27.7	22.8	54.8	0.0	0.0	0.0	7.6	19.6	24.0	17.3	54.8
1973	14.8	21.7	11.6	70.8	5.1	0.0	0.0	0.0	10.1	13.6	38.8	26.1	70.8
1974	23.4	18.2	10.7	17.7	49.3	4.3	0.0	0.0	10.1	34.0	14.3	31.6	49.3
1975	17.4	33.1	45.1	29.0	5.3	0.0	2.9	0.0	5.2	14.3	18.3	12.1	45.1
1976	43.2	36.4	21.1	15.7	9.9	0.0	3.7	0.0	9.7	19.8	22.0	18.7	43.2
1977	27.6	33.8	32.5	32.6	47.4	21.2	0.0	0.0	14.3	19.6	26.5	22.7	47.4
1978	44.8	19.1	10.0	44.0	8.0	0.0	0.0	0.0	23.0	21.8	40.0	39.6	44.8
1979	33.8	19.9	13.3	21.4	20.1	0.0	0.0	0.0	7.3	18.5	47.0	36.2	47.0
1980	35.0	30.9	28.4	40.0	8.2	18.0	0.0	6.1	10.1	21.7	50.8	29.0	50.8
1981	29.0	33.6	65.2	55.0	4.1	0.0	0.0	3.5	9.7	19.8	48.0	16.7	65.2
1982	21.2	20.0	35.8	42.9	26.0	0.0	0.0	0.0	9.5	30.0	34.5	24.3	42.9
1983	22.5	38.0	21.1	28.3	17.5	0.0	3.7	0.0	0.0	26.5	34.0	30.0	38.0
1984	13.3	36.4	35.5	15.7	9.9	0.0	0.0	11.2	12.3	16.9	22.0	18.7	36.4
1985	27.6	4.9	41.4	38.0	34.4	0.0	10.2	0.2	15.5	19.0	35.1	40.0	41.4
1986	20.9	8.2	10.7	23.6	37.4	12.1	0.0	0.0	19.0	34.9	30.0	31.6	37.4
1987	20.1	35.1	35.0	29.6	5.3	0.0	2.9	0.0	14.5	14.7	37.3	23.8	37.3
1988	25.7	40.4	58.6	31.5	3.6	1.1	9.1	0.0		36.1	40.0	39.5	58.6
1989	31.0	34.0	16.4	42.4	4.0	0.8	0.0	1.8	6.3	28.1	30.0	31.7	42.4
1990	23.1	50.0	28.7	13.5	60.6	0.0	0.0	0.0	7.3	22.2	32.5	23.2	60.6
1991	25.5	23.4	27.4	23.0	31.3	0.0	0.0	7.5	8.9	19.4	25.8	25.8	31.3
1992	28.7	27.5	25.6	13.0	25.9	0.0	0.0	0.0	30.0	20.2	35.0	10.5	35.0
1993	17.0	6.0	30.7	31.1	30.1	0.0	0.0	0.0	5.0	27.0	31.5	11.7	31.5
1994		2.6	10.5	24.0	17.4	0.0	0.0			9.1	35.1	29.4	35.1
1995	29.8	39.4	43.2	13.8	26.6	0.0	0.0	26.0	36.9	14.6	42.4	37.1	43.2
1996	39.8	10.0	43.1	36.5	31.1	0.2	0.0	0.0	5.1	18.0	28.4	25.4	43.1
1997	39.5	18.7	29.3	38.0	23.6	1.7	0.0	2.5	1.3	38.5	24.1	25.3	39.5
1998	26.2	30.9	50.2	43.6	20.1	0.0	0.0	0.0	29.0	21.8	25.7	33.3	50.2
1999	32.2	31.4	49.4	11.6	17.6	8.6	0.7	0.0		42.8		45.0	49.4
2000	53.2	29.9	22.0	31.4	19.0	0.0	0.0	0.0	30.0	21.7	44.2	41.8	53.2
2001	11.2	45.3	25.8		39.2								45.3
2002													0.0

5. QUANTITE D'EAU ENREGISTREE EN 60 minutes.

STATION DE KIN-BINZA

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D	MAX. ANN
1977	38.9	39.8	53.6	37.0	56.1	23.2	0.0	0.0	15.2	23.1	41.6	27.6	56.1
1978	69.2	26.5	12.6	50.3	12.9	0.0	0.0	0.0	26.9	25.1	65.7	56.5	69.2
1979	40.9	37.2	17.0	27.0	21.0	0.0	0.0	0.0	9.6	22.8	63.0	41.7	63.0
1980	45.5	35.0	28.6	53.7	8.5	0.0	0.0	13.7	10.7	38.1	58.3	47.5	58.3
1981	45.8	45.0	82.7	60.0	0.0	0.0	0.0	4.8	11.3	27.4	53.9	17.2	82.7
1982	28.6	22.1	50.3	53.7	26.8	0.0	0.0	0.0	10.2	52.0	47.2	31.8	53.7
1983	30.0	65.0	22.4	33.0	17.7	0.0	4.0	0.0	0.0	29.3	39.6	20.2	65.0
1984	14.7	39.8	60.1	18.3	10.5	0.0	0.0	11.5	16.7	22.0	30.7	15.4	60.1
1985	33.2	1.0	63.7	76.5	50.5	0.0	13.5	0.4	23.7	23.2	44.1	50.3	76.5
1986	28.5	9.6	10.8	32.8	62.0	20.5	0.0	0.0	23.1	67.0	45.1	36.7	67.0
1987	24.0	49.5	48.1	39.8	7.1	0.0	0.0	0.0	17.3	16.4	41.4	32.3	49.5
1988	28.5	52.9	74.0	34.5	3.8	1.3	10.5	0.0		44.6	60.0	62.5	74.0
1989	48.5	45.5	22.5	57.0	4.4	1.0	0.0	2.5	7.9	38.5	41.5	35.9	57.0
1990	29.7	61.7	24.0	25.0	83.6	0.0	0.0	0.0	10.0	31.5	57.7	37.0	83.6
1991	44.3	38.4	31.0	28.0	40.8	0.0	0.0	7.6	12.1	30.8	28.4	28.4	44.3
1992	37.7	36.3	37.6	18.0	24.5	0.0	0.0	0.0	32.0	22.8	56.0	13.8	56.0
1993	32.0	7.1	33.2	33.9	35.2	0.0	0.0	0.0	5.5	45.0	34.2	15.5	45.0
1994		3.4	14.3	28.3	19.4	0.0	0.0			13.3	65.4	42.1	65.4
1995	35.1	40.5	54.5	18.1	32.7	0.0	0.0	36.8	49.8	16.8	47.2	43.2	54.5
1996	60.9	16.0	57.5	40.3	33.7	0.2	0.0	0.0	6.3	24.4	31.0	26.5	60.9
1997	42.3	21.1	35.4	50.5	34.7	2.3	0.0	3.6	1.4	49.3	42.2	48.7	50.5
1998	41.0	35.8	62.4	67.9	26.3	0.0	0.0	0.0	31.6	43.7	29.7	49.1	67.9
1999	40.7	35.2	58.8	11.8	22.8	10.6	0.8	0.0		52.8		46.9	58.8
2000	69.3	52.8	26.1	40.0	26.0	0.0	0.0	0.0	52.7	27.9	53.5	54.2	69.3
2001	26.4	53.0	37.1		77.4								
2002													

STATION PLUVIOMETRIQUE DE NDJILI :

STATION DE KINSHASA - NDJILI

Longitude : 15° 26' E
Latitude : 04° 23' S
Altitude : 311 m
Indicatif : 64210

INTENSITES DES PLUIES en mm

1. INTENSITES MAXIMALE en 5 Minutes.

STATION DE KINSHASA - NDJILI

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D	MAX. ANN.
1977	6.0	5.4	14.0	5.5	5.5	2.5	0.0	0.0	3.5	9.1	10.0		14.0
1978	6.0	1.8	7.5	5.5	13.5	0.0	0.0	0.0	1.3	4.1	15.0	5.5	15.0
1979	6.5	14.0	14.0	5.5		0.0	0.0	0.0	0.5	5.0	7.0	6.0	14.0
1980	10.0	15.5	10.0	7.5	10.0	0.0	0.0	0.0	2.5	3.1	20.6	5.3	20.6
1981	5.3	5.8	6.6	5.0	11.5	0.0	0.0	1.0	0.4	4.0	8.0	8.5	11.5
1982	6.0	7.0	9.5	2.5	9.2	0.0	0.0	0.0	3.5	3.7	10.8	3.1	10.8
1983	4.3	6.0	8.3	9.9	8.8	0.0	0.0	0.0	0.0	15.5	10.0	9.8	15.5
1984	9.8	19.6	9.7	5.4	9.8	2.4	2.0	0.0	3.6	10.0	11.5	8.0	19.6
1985	10.0	6.0	10.0	10.0	6.1	0.0	0.0	0.0	1.3	10.0	9.7	9.4	10.0
1986	3.5	6.0	10.9	10.0	8.0	4.4	0.0	0.0	5.5	7.0	7.7	7.9	10.9
1987	10.3	10.0	10.2	10.2	6.0	0.0	0.0	0.0	2.5	10.1	10.1	10.4	10.4
1988	3.8	12.9	6.5	10.0	2.0	0.0	2.2	0.0	0.0	20.0		10.0	20.0
1989	6.4	20.0	10.0	10.0	11.0	0.0	0.0	0.0	11.4	10.0	10.0	10.0	20.0
1990	9.0	9.9	8.0	3.0	10.0	2.5	0.0	0.0	5.0	4.7	10.1	10.0	10.1
1991	7.4	2.8	5.0	6.6	9.9	0.0	0.6	0.0	0.0	6.7	7.1	5.0	9.9
1992	10.0	15.0	6.2			0.0	0.0	0.0	3.4		8.0	6.0	15.0
1993	10.1	6.0	8.0	2.5	10.0	0.0	0.0	0.0	1.0	5.1	3.5	8.0	10.1

INTENSITES MAXIMALES en 10 Minutes.

STATION DE KINSHASA - NDJILI

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D	MAX. ANN.
1977	11.4	7.4	24.0	10.7	16.5	4.5	0.0	0.0	7.5	14.1	20.5		24.0
1978	10.5	2.0	10.0	9.0	21.7	0.0	0.0	0.0	3.2	6.1	20.0	10.0	21.7
1979	11.3	23.0	20.5	10.0		0.0	0.0	0.0	1.0	10.0	11.8	7.0	23.0
1980	11.0	20.0	15.4	19.5	18.0	0.0	0.0	0.0	5.5	0.0	26.3	8.8	26.3
1981	9.8	10.8	10.4	10.0	11.0	0.0	0.0	1.4	0.9	5.4	15.4	14.2	15.4
1982	11.3	10.0	10.1	3.4	20.8	0.0	0.0	0.0	6.5	8.7	14.3	4.3	20.8
1983	5.5	15.4	9.8	19.7	11.0	0.0	0.0	0.0		19.5	20.0	11.8	20.0
1984	29.4	29.4	19.4	7.5	13.9	2.9	2.0	0.0	8.6	10.7	21.5	10.3	29.4
1985	17.2	9.9	15.9	20.0	9.3	0.0	0.0	0.0	1.6	20.0	15.6	16.4	20.0
1986	8.0	8.8	14.9	13.5	13.0	6.4	0.0	0.0	9.3	12.0	13.1	11.5	14.9
1987	13.8	21.7	19.3	20.0	6.9	0.0	0.0	0.0	3.7	16.4	14.3	13.0	21.7
1988	10.1	17.9	10.0	15.9	4.2	0.0	3.7	0.0	0.0	30.0		11.2	30.0
1989	10.0	27.0	12.0	15.0	21.0	0.0	0.0	0.0	15.0	15.0	12.0	15.3	27.0
1990	14.8	17.2	10.0	6.0	13.3	2.6	0.0	0.0	7.5	8.2	15.0	14.6	17.2
1991	13.0	4.2	10.0	12.3	20.4	0.0	1.1	0.0	0.0	9.9	9.6	9.5	20.4
1992	24.6	22.0	13.6			0.0	0.0	0.0	3.9		12.5	10.0	24.6
1993	20.0	7.0	14.0	5.2	20.2	0.0	0.0	0.0	2.3	10.1	7.0	17.2	20.2

3. INTENSITES MAXIMALES en 20 Minutes.

STATION DE KINSHASA - NDJILI

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D	MAX. ANN.
1977	18.0	11.4	37.0	14.0	21.0	8.5	0.0	0.0	11.8	25.1	38.5		38.5
1978	21.0	3.3	16.6	16.0	25.2	0.0	0.0	0.0	5.7	10.0	27.6	22.5	27.6
1979	19.4	35.5	28.4	14.5		0.0	0.0	0.0	2.1	14.0	16.8	14.0	35.5
1980	20.1	55.3	16.0	30.2	27.0	0.0	0.0	0.0	7.6	10.8	36.7	15.1	55.3
1981	18.1	25.0	18.5	15.3	11.6	0.0	0.0	1.6	1.6	5.8	30.3	25.6	30.3
1982	19.5	18.0	25.0	4.0	23.9	0.0	0.0	0.0	10.7	10.0	28.5	8.5	28.5
1983	6.4	18.9	13.8	29.5	11.8	0.0	0.0		0.0	29.2	30.0	19.6	30.0
1984	34.4	43.1	29.1	10.5	11.6	3.7	3.1	0.0	13.1	15.5	22.8	17.9	43.1
1985	20.0	15.7	22.0	34.5	11.3	0.0	0.0	0.0	2.3	40.0	18.4	23.8	40.0
1986	11.5	9.5	22.3	26.5	20.7	7.2	6.9	0.0	14.1	21.0	20.1	22.5	26.5
1987	20.7	27.4	23.4	20.3	9.4	0.0	0.0	0.0	6.1	16.6	30.6	22.4	30.6
1988	20.2	30.0	14.5	22.8	5.8	0.0	6.0	0.0	0.0	47.5		12.5	47.5
1989	19.0	32.1	20.0	22.1	32.0				18.3	20.0	18.4	26.2	32.1
1990	19.0	21.3	17.0	10.5	24.0	2.7	0.0	0.0	9.9	14.2	22.1	24.7	24.7
1991	19.2	5.3	15.0	22.6	33.5	0.0	1.3	0.0	0.0	19.4	16.7	20.0	33.5
1992	33.8	32.1	27.6			0.0	0.0	0.0	3.9		22.6	16.0	33.8
1993	33.0	12.5	27.0	13.7	31.4	0.0	0.0	0.0	4.3	17.0	12.1	27.2	33.0

4. INTENSITES MAXIMALES en 30 Minutes.

STATION DE KINSHASA - NDJILI

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D	MAX. ANN
1977	22.0	13.4	55.0	18.7	21.5	9.0	0.0	0.0	14.3	31.1	56.5		56.5
1978	29.2	3.5	18.5	18.6	34.2	0.0	0.0	0.0	7.2	13.0	44.0	29.0	44.0
1979	24.4	44.5	36.5	18.0		0.0	0.0	0.0	2.4	26.0	20.1	20.0	44.5
1980	22.0	62.7	16.5	39.5	32.3	0.0	0.0	0.0	8.5	12.4	41.2	20.3	62.7
1981	21.4	32.5	24.7	20.8	11.6	0.0	0.0	2.2	2.9	5.9	38.0	27.7	38.0
1982	20.5	27.5	32.8	6.2	29.4	0.0	0.0	0.0	11.6	10.7	37.7	9.3	37.7
1983	7.8	20.4	18.3	39.3	11.9	0.0	0.0		0.0	34.2	32.9	21.6	39.3
1984	44.3	48.9	38.8	11.3	21.9	4.0	3.1	0.0	13.6	17.8	23.2	20.5	48.9
1985	20.5	17.7	22.2	35.1	13.3	0.0	0.0	0.0	2.7	50.0	18.6	33.6	50.0
1986	14.5	10.1	23.9	34.0	25.3	7.3	6.9	0.0	18.1	29.3	26.1	34.5	34.5
1987	23.6	37.0	33.6	20.3	14.0	0.0	0.0	0.0	6.6	16.7	40.8	27.3	40.8
1988	21.7	36.0	17.0	25.9	5.9	0.0	8.2	0.0	0.0	61.4		12.5	61.4
1989	20.1	34.1	28.0	23.6	37.5	0.0	0.0	0.0	19.3	25.8	22.0	29.2	37.5
1990	30.5	22.2	17.2	12.0	38.0	3.8	0.0	0.0	10.8	18.4	36.8	39.2	39.2
1991	30.3	6.5	17.3	29.1	51.4	0.0	1.3	0.0	0.0	24.3	21.0	22.8	51.4
1992	35.6	41.3	40.1			0.0	0.0	0.0	3.9		25.6	18.3	41.3
1993	45.1	23.0	37.0	23.1	40.1	0.0	0.0	0.0	5.7	25.2	17.5	35.3	45.1

5. INTENSITES MAXIMALES en 60 Minutes.

STATION DE KINSHASA - NDJILI

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D	MAX. ANN.
1977	34.5	15.4	75.0	23.7	22.0	11.5	0.0	0.0	15.6	39.6	77.8		77.8
1978	33.2	7.5	19.2	25.5	35.2	0.0	0.0	0.0	8.4	16.5	64.0	43.8	64.0
1979	31.2	55.5	52.5	30.2		0.0	0.0	0.0	2.6	29.8	28.8	23.2	55.5
1980	22.5	23.1	19.6	57.2	36.2	0.0	0.0	0.0	10.0	13.5	45.1	26.0	57.2
1981	21.5	35.7	43.9	31.8	11.6	0.0	0.0	4.4	5.4	6.0	57.3	31.1	57.3
1982	22.8	35.8	44.3	7.7	29.4	0.0	0.0	0.0	12.3	12.7	38.2	9.3	44.3
1983	10.5	26.0	19.6	68.7	12.0	0.0	0.0		0.0	46.9	33.4	31.8	68.7
1984	59.0	71.5	63.7	12.6	22.1	4.0	3.1	0.0	20.3	24.0	24.0	21.8	71.5
1985	20.9	19.2	22.3	42.9	14.0	0.0	0.0	0.0	2.9	61.8	18.7	47.1	61.8
1986	20.7	12.3	24.3	39.4	36.1	7.4	0.0	0.0	34.1	53.5	39.7	55.2	55.2
1987	24.4	58.2	24.4	20.3	15.1	0.0	0.0	0.0	8.3	17.7	70.6	28.3	70.6
1988	33.4	51.1	23.4	29.3	5.9	0.0	9.6	0.0	0.0	80.0		12.7	80.0
1989	29.3	66.0	43.1	24.0	39.3	0.0	0.0	0.0	20.2	25.9	43.0	31.1	66.0
1990	34.9	22.8	20.7	24.9	48.5	4.0	0.0	0.0	11.3	26.8	53.8	42.3	53.8
1991	38.8	11.3	18.9	32.6	75.1	0.0	1.4	0.0	0.0	33.2	23.0	24.7	75.1
1992	40.2	54.8	57.4			0.0	0.0	0.0	3.9		33.9	29.0	57.4
1993	65.2	37.2	60.0	26.3	53.6	0.0	0.0	0.0	7.5	33.2	20.4	45.3	65.2

STATION PLUVIOMETRIQUE DE NDOLO

STATION DE KIN-NDOLO

INTENSITE MAXIMALE DES PLUIES en mm.

Longitude :
Latitude :
Altitude :
Indicatif : 64211

1, QUANTITE D'EAU ENREGISTREES EN 5 minutes.

STATION DE KIN-NDOLO

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D
2004								0.0	5.0	7.0	6.0	6.2
2005	1.8	4.2	14.9	10.2	8.2	0.8	0.0	0.0	3.7	8.0	10.5	7.5
2006	5.5	7.5	9.0	8.2	5.6	0.8	0.0	0.7	1.0	8.0	4.4	7.5
2007	7.9	4.2	8.0								6.2	4.5
2008	1.7	0.9	2.8									

2, QUANTITE D'EAU ENREGISTREE EN 10 minutes.

STATION DE KIN-NDOLO

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D
2004								0.0	9.0	15.9	10.0	12.9
2005	2.8	7.5	30.7	15.7	16.4	0.9	0.0	0.0	5.3	16.1	20.5	13.4
2006	7.9	13.8	16.4	15.0	8.9	0.9	0.0	0.9	1.4	15.7	7.0	19.5
2007	12.2	11.1	16.8								14.7	6.1
2008	2.3	2.1	3.7									

3, QUANTITE D'EAU ENREGISTREES EN 20 minutes.

STATION DE KIN-NDOLO

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D
2004								0.0	11.0	31.5	16.3	16.5
2005	4.5	11.6	48.8	30.5	21.9	1.8	0.0	0.0	12.8	24.8	25.5	21.3
2006	14.0	21.3	23.8	34.9	17.5	0.9	0.0	1.1	1.6	30.1	10.8	28.4
2007	15.0	15.4	23.4								30.6	15.5
2008	4.2	3.6	10.0									

4. INTENSITES MAXIMALES en 30 Minutes.

STATION DE KIN-NDOLO

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D
2004								0.0	12.6	42.0	21.7	27.6
2005	6.2	12.9	64.7	43.1	24.4	2.2	0.0	0.0	17.3	29.9	30.7	24.9
2006	15.6	26.8	25.1	49.0	28.1	0.9	0.0	1.6	2.5	36.8	12.5	30.4
2007	24.5	18.8	28.5								43.4	16.7
2008	5.1	3.7	12.7									

5. INTENSITES MAXIMALES en 60 Minutes.

STATION DE KIN-NDOLO

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D
2004								0.0	12.7	54.7	25.0	31.2
2005	11.0	13.2	67.5	64.2	30.0	3.3	0.0	0.0	27.4	39.5	35.3	41.4
2006	22.0	27.3	28.0	71.7	37.7	0.9	0.0	1.6	4.5	55.7	20.9	61.8
2007	26.6	21.2	30.3								45.1	22.6
2008	6.4	3.9	13.5									

STATION PLUVIOMETRIQUE DE CRREBaC/MONT-AMBA

STATION DE MONT AMBA CREEBAC

INTENSITE MAXIMALE DES PLUIES en mm.

Longitude :
Latitude :
Altitude :
Indicatif :

1, QUANTITE D'EAU ENREGISTREES EN 5 minutes.

STATION DE MONT AMBA CREEBAC

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D
2022		3.2	9.3	5.8	4.1	0.1	0	0	0.1	11.6	4.7	10.1
2023	7	7.5	12.3	4.2	4.3	0.4	0	0.5	2.5	5.5	11.6	6.5
2024	8.7	4.7	6.1	7	5.2	0	0	0	4.1	5.6	12.7	3.3
2025	6.1	4.7		7	9.5	9.8	0	0	4.9	2.5	15.2	

2, QUANTITE D'EAU ENREGISTREE EN 10 minutes.

STATION MONT AMBA CREEBAC

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D
2022		6.4	18.6	11.6	8.2	0.2	0	0	0.2	23.2	9.4	20.2
2023	14	15	24.6	8.4	8.6	0.8	0	1	5	11	23.2	13
2024	17.4	9.4	12.2	14	10.4	0	0	0	8.2	11.2	25.4	6.6
2025	12.2	9.4		14	19	19.6	0	0	9.8	5	30.4	

3, QUANTITE D'EAU ENREGISTREES EN 20 minutes.

STATION DE MONT AMBA CREEBAC

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D
2022		9.8	36.6	17	14.4	0.2	0	0	0.2	42.6	16.2	32.6
2023	26	28.4	41.8	13.6	8.8	1.4	0	1.6	5	20	44.6	20.2
2024	31.8	16.4	21	20.8	15.4	0	0	0	9.6	21.8	40.8	12.4
2025	22.8	18.8		23	37	35	0	0	16.4	9.4	56.4	

4. INTENSITES MAXIMALES en 30 Minutes.

STATION DE MONT AMBA CREEBAC

ANNEES	J	F	M	A	M	J	J	A	S	O	N	D
2022		13	47	20.2	18.6	0.2	0	0	0.2	61.2	22.8	43
2023	36.8	36.2	54.4	14.8	8.8	2	0	2.4	5	29	65.4	28.2
2024	37.4	22.2	28.6	25.8	16.8	0	0	0	10	32	52.8	15.8
2025	30	19.2		28.2	51.2	45.6	0	0	18.2	11	72.2	

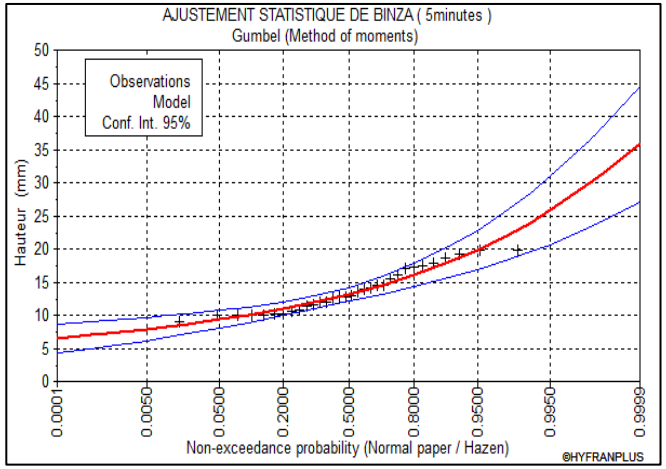
5. INTENSITES MAXIMALES en 60 Minutes.

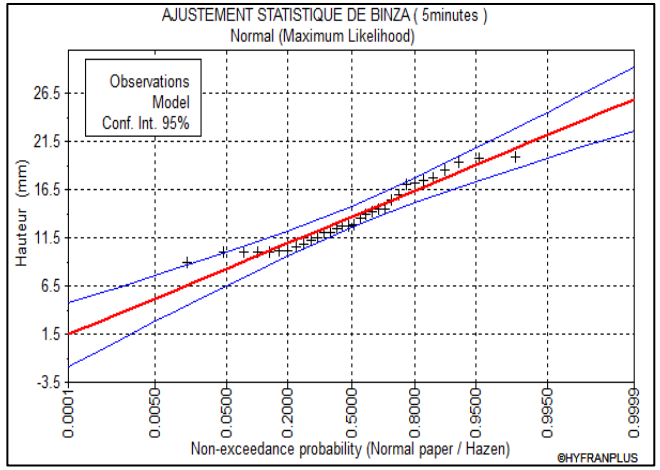
STATION MONT AMBA CREEBAC

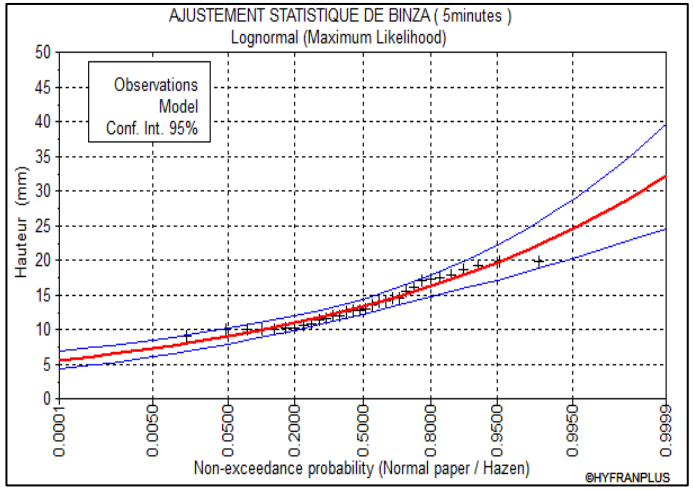
ANNEES	J	F	M	A	M	J	J	A	S	O	N	D
2022		13.8	74.2	32	25.2	0.2	0	0	0.4	96	32.2	50.2
2023	43.6	53.8	66	16.4	9.2	3.4	0	2.8	5	33.8	87	42.2
2024	44.4	23.6	38.2	29	17	0	0	0	10.2	44.4	76	22.8
2025	31.2	25.8		40.2	73	70.8	0	0	19.6	16.4	81.6	

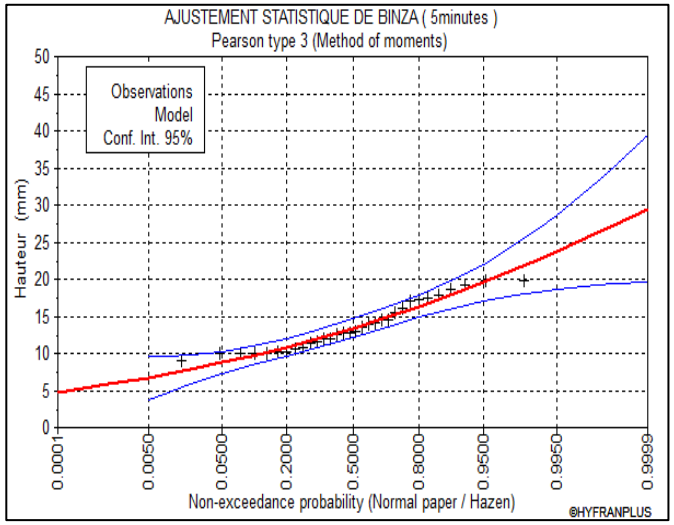
III GRAPHIQUE D'AJUSTEMENTS STATISTIQUE DES LOIS (STATION BINZA)

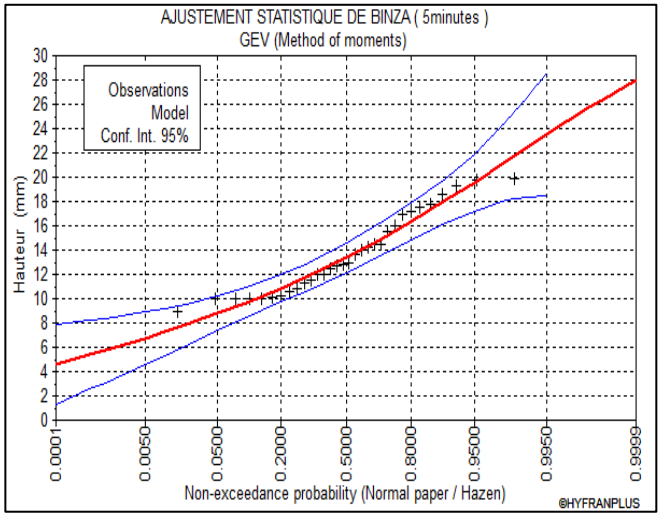
5minutes



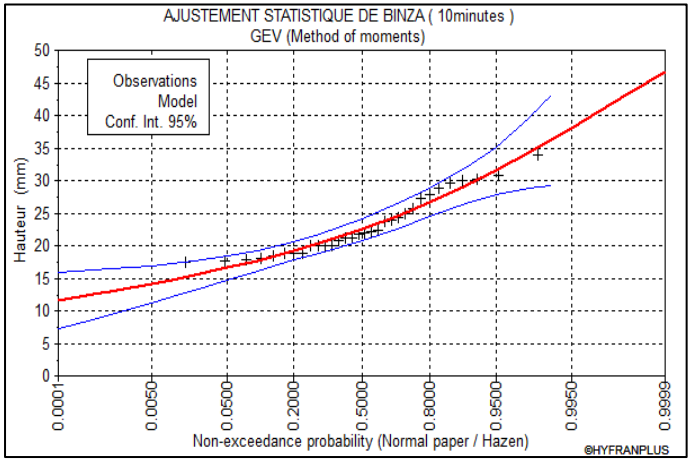


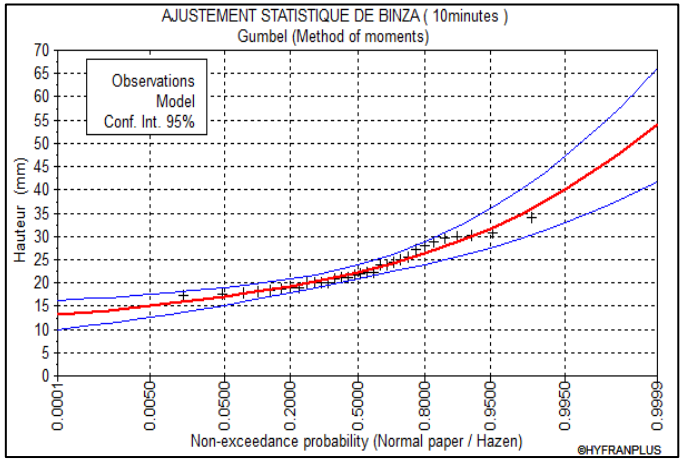


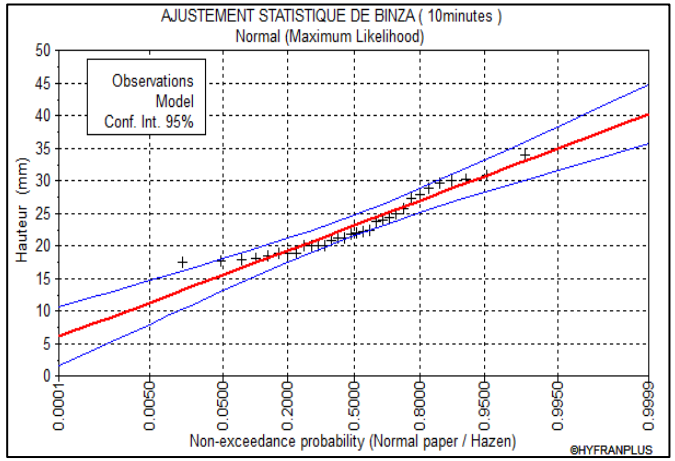


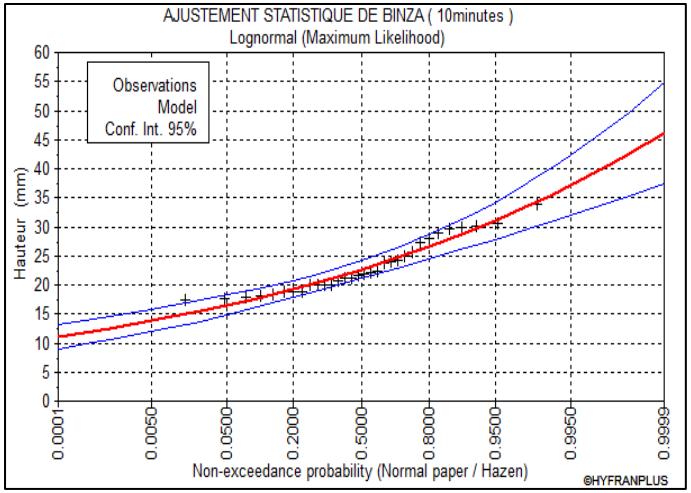


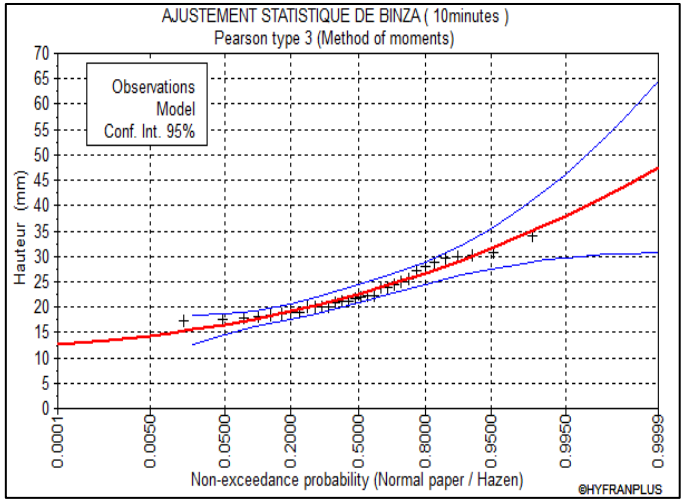
10 minutes


AJUSTEMENT STATISTIQUE DE BINZA (10minutes)
GEV (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

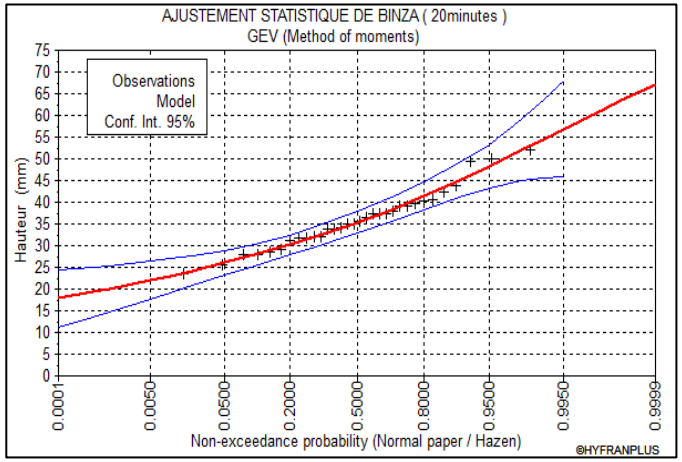

AJUSTEMENT STATISTIQUE DE BINZA (10minutes)
Gumbel (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

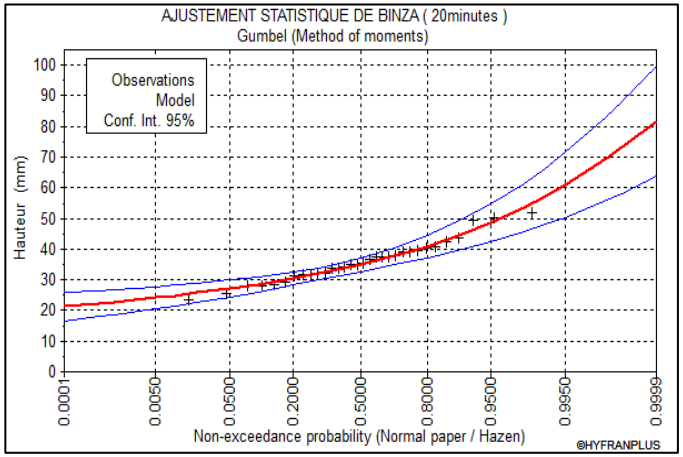

AJUSTEMENT STATISTIQUE DE BINZA (10minutes)
Normal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

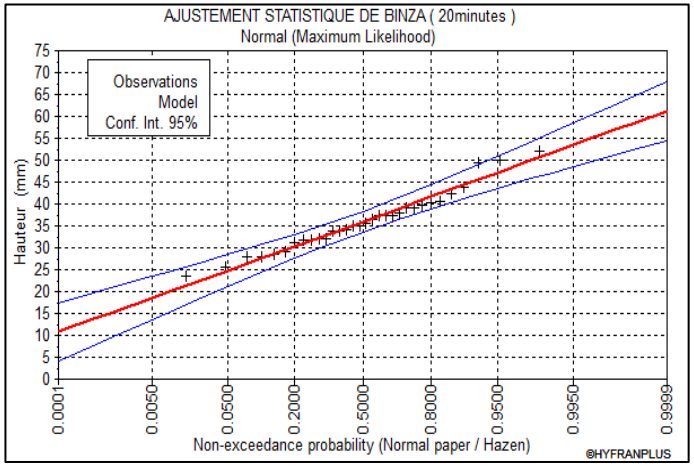

AJUSTEMENT STATISTIQUE DE BINZA (10minutes)
Lognormal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

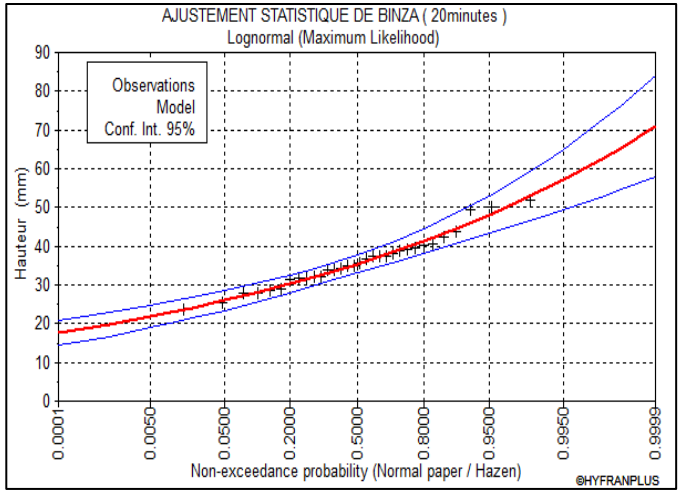

AJUSTEMENT STATISTIQUE DE BINZA (10minutes)
Pearson type 3 (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

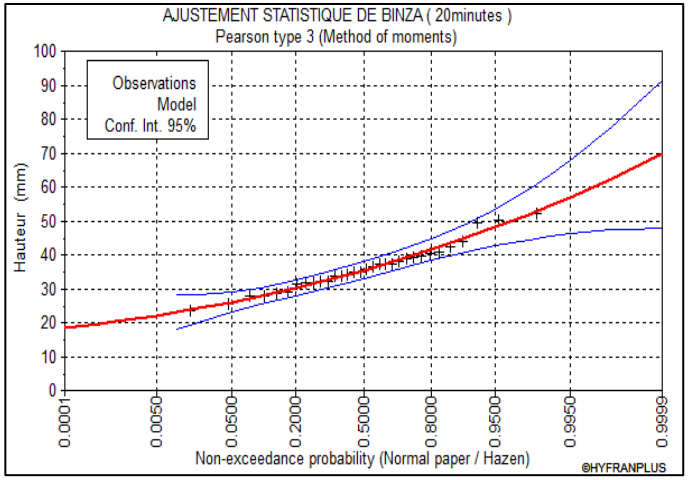
20 MINUTES


AJUSTEMENT STATISTIQUE DE BINZA (20minutes)
GEV (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

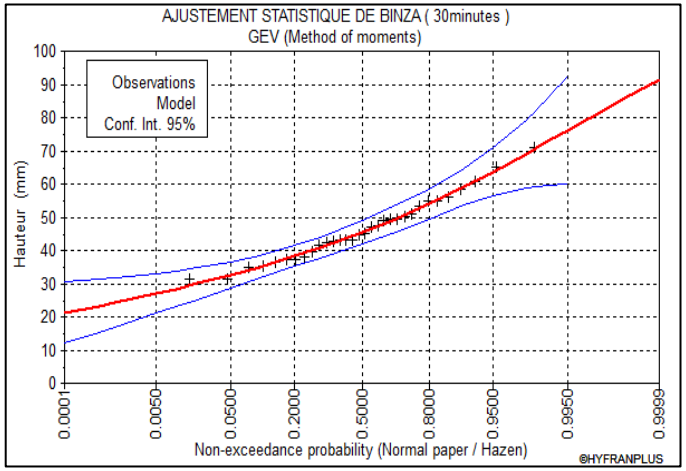

AJUSTEMENT STATISTIQUE DE BINZA (20minutes)
Gumbel (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

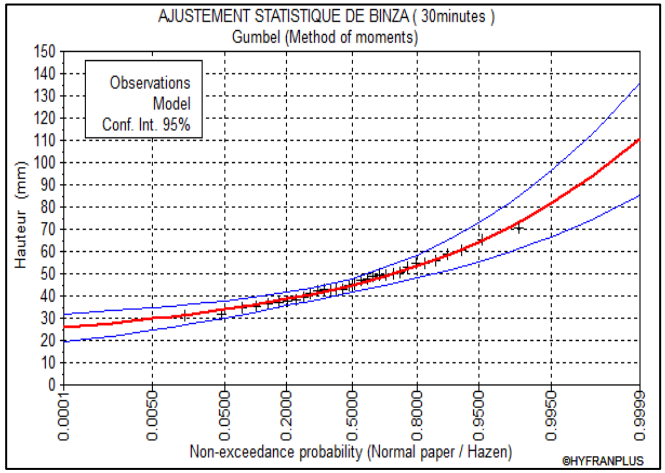

AJUSTEMENT STATISTIQUE DE BINZA (20minutes)
Normal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

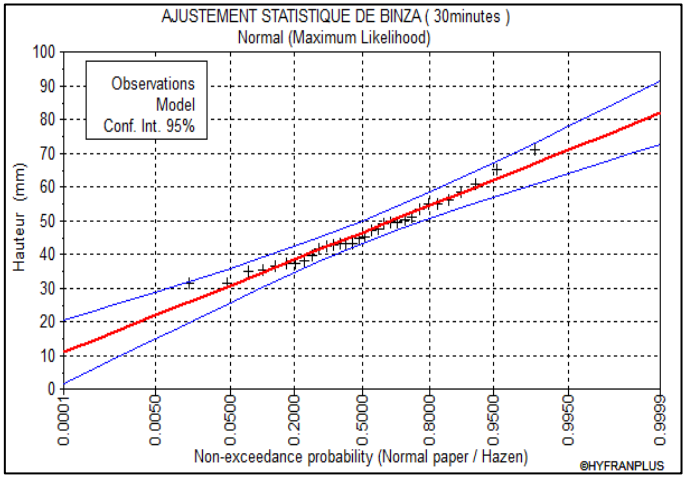

AJUSTEMENT STATISTIQUE DE BINZA (20minutes)
Lognormal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

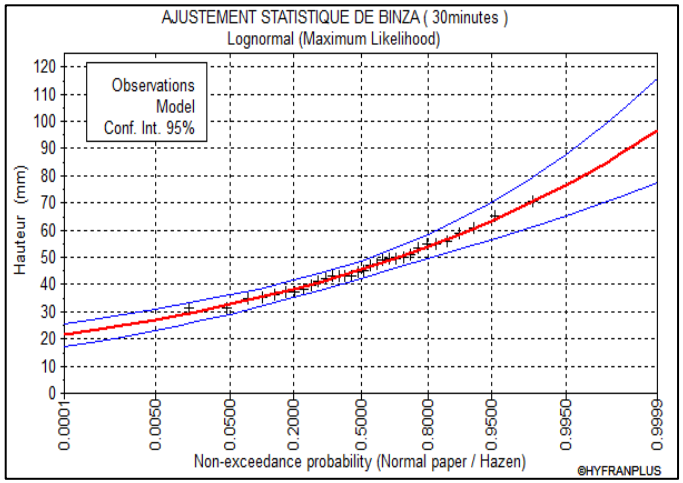

AJUSTEMENT STATISTIQUE DE BINZA (20minutes)
Pearson type 3 (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

30 MINUTES


AJUSTEMENT STATISTIQUE DE BINZA (30minutes)
GEV (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE BINZA (30minutes)
Gumbel (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE BINZA (30minutes)
Normal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE BINZA (30minutes)
Lognormal (Maximum Likelihood)
Observations
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Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
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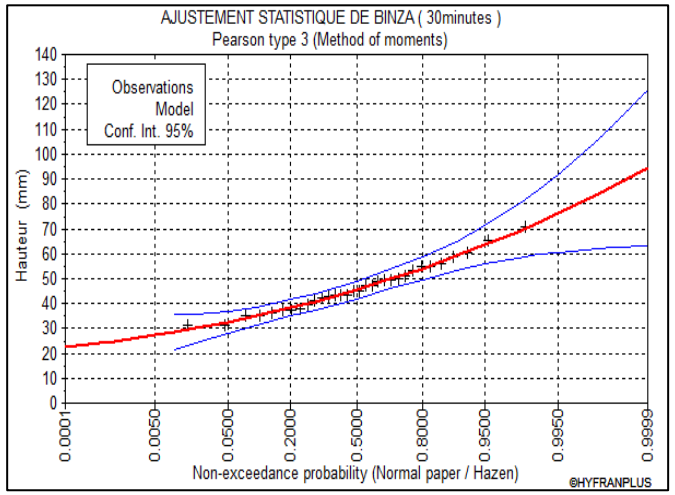
AJUSTEMENT STATISTIQUE DE BINZA (30minutes)
Pearson type 3 (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
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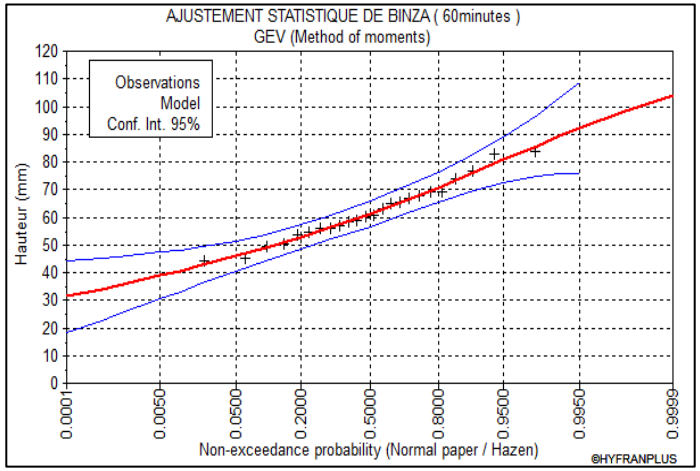
60 MINUTES


AJUSTEMENT STATISTIQUE DE BINZA (60minutes)
GEV (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
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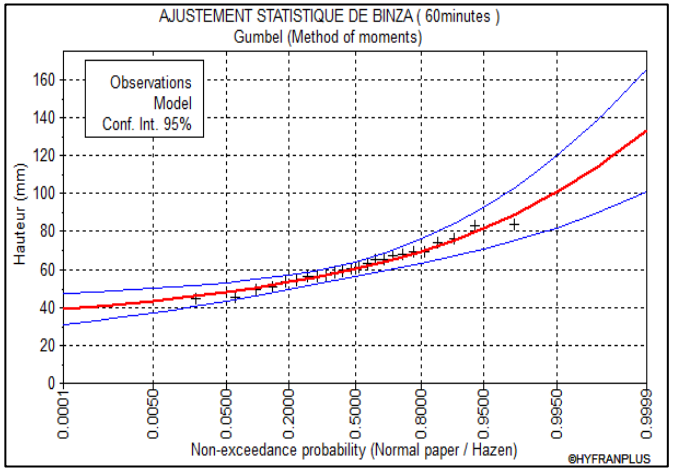
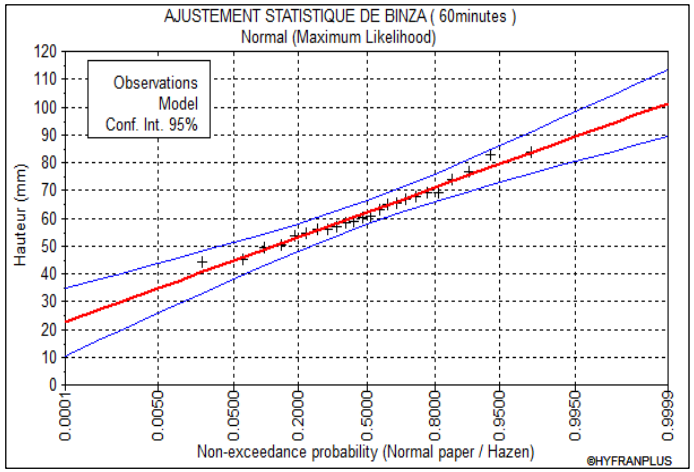

AJUSTEMENT STATISTIQUE DE BINZA (60minutes)
Gumbel (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE BINZA (60minutes)
Normal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
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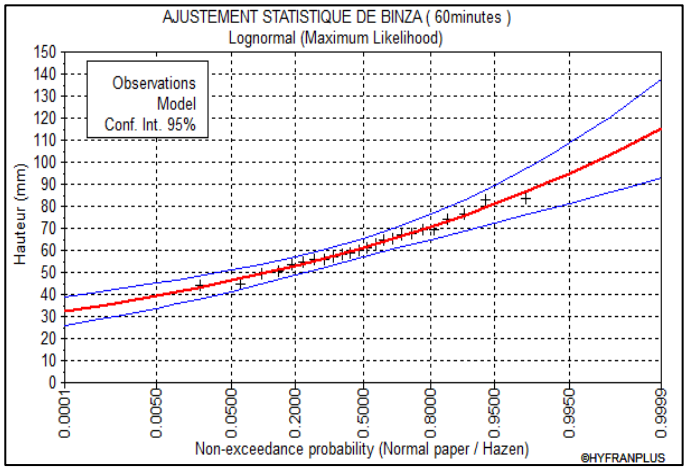
AJUSTEMENT STATISTIQUE DE BINZA (60minutes)
Lognormal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
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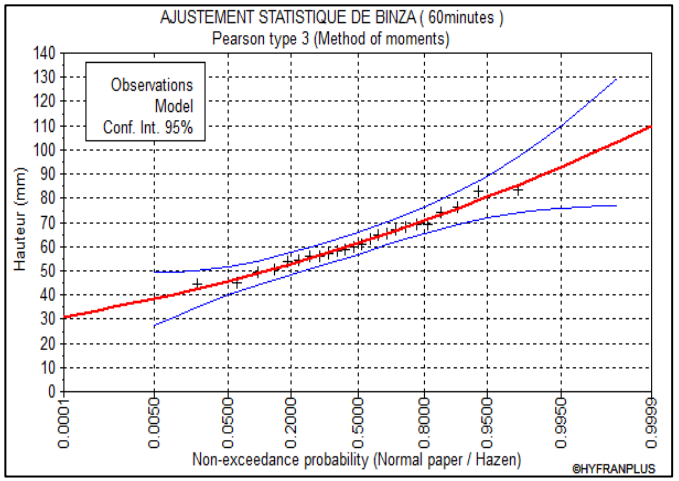
AJUSTEMENT STATISTIQUE DE BINZA (60minutes)
Pearson type 3 (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
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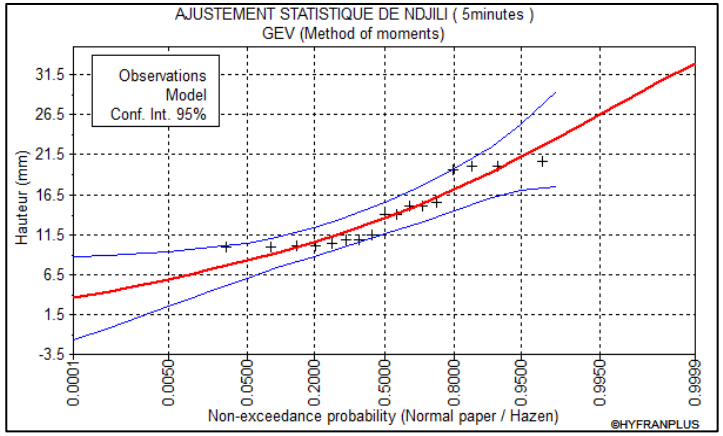
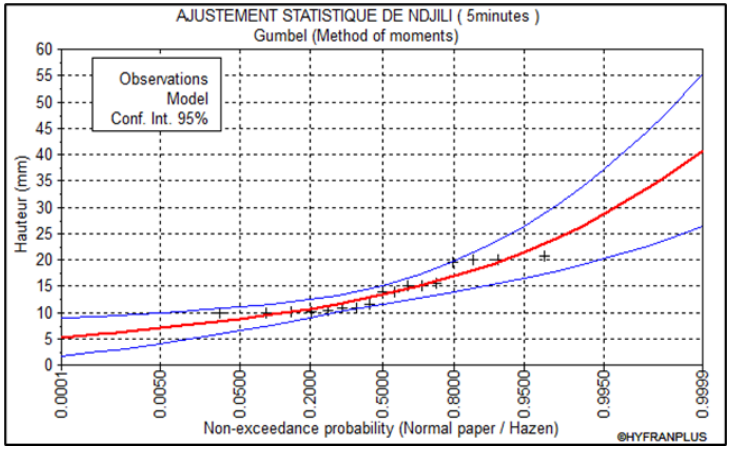
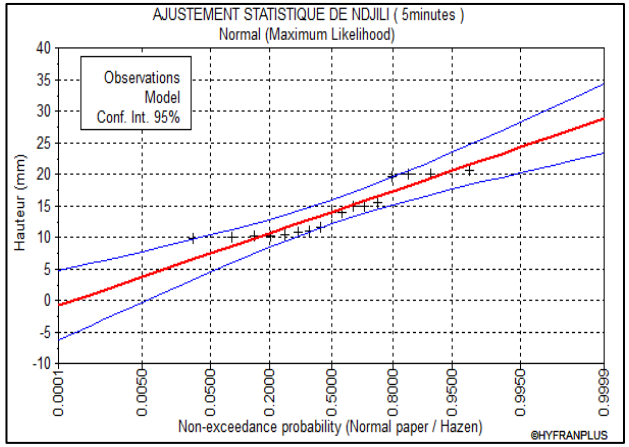
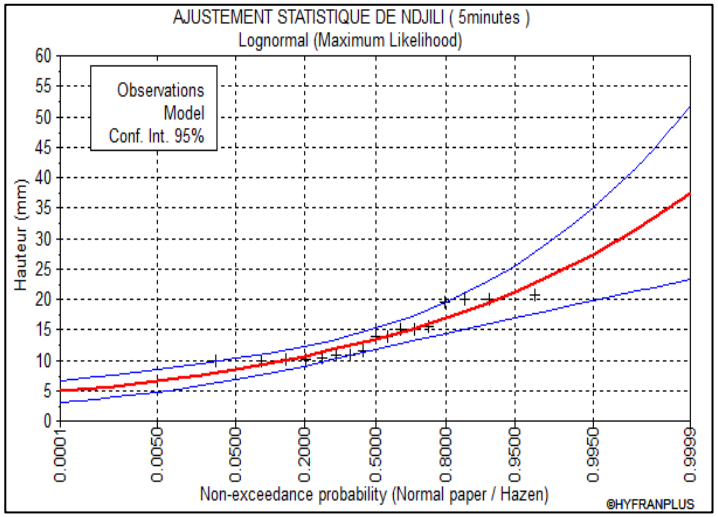
GRAPHIQUE D'AJUSTEMENT DES LOIS (STATION NDJILI)

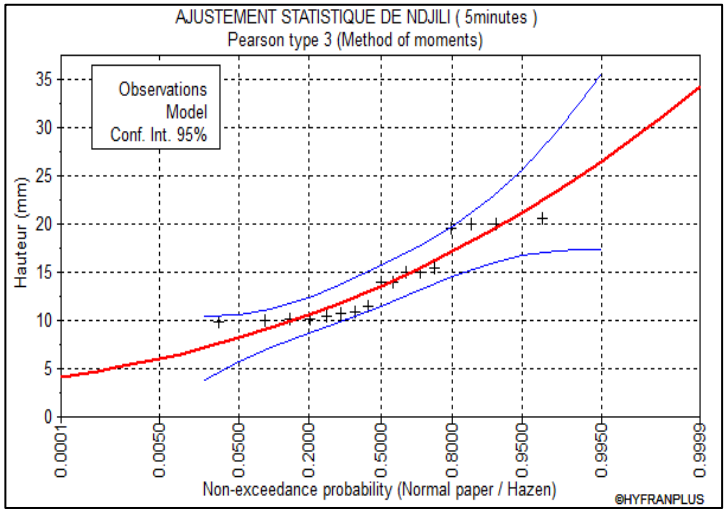
5minutes



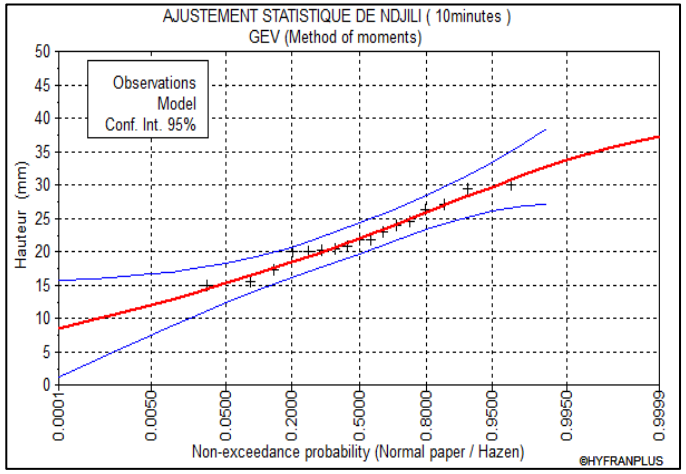


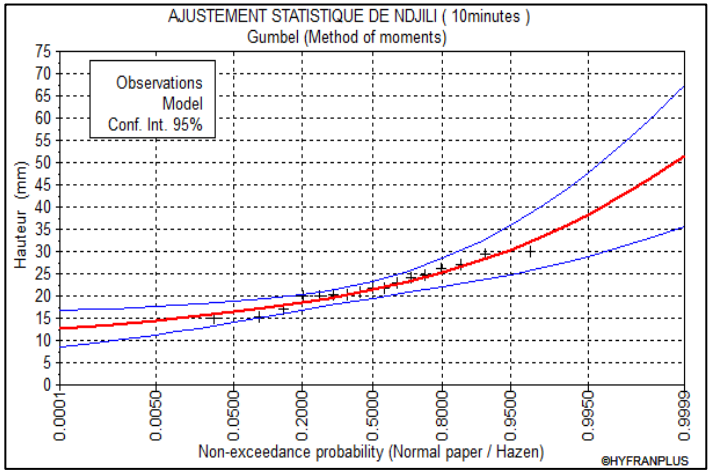


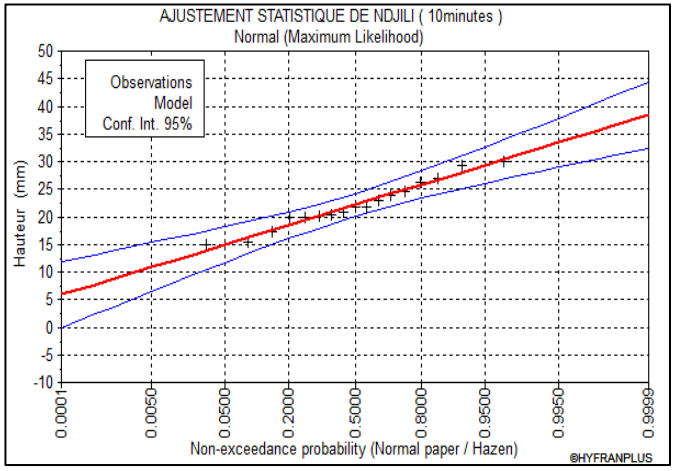


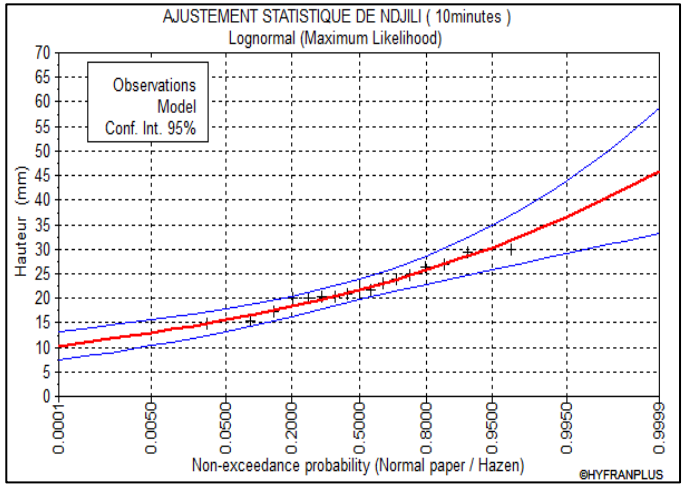


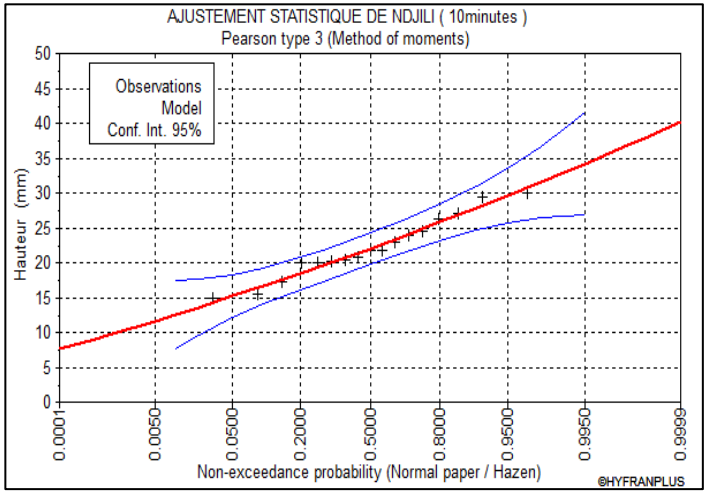
10 MINUTES


AJUSTEMENT STATISTIQUE DE NDJILI (10minutes)
GEV (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE NDJILI (10minutes)
Gumbel (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE NDJILI (10minutes)
Normal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE NDJILI (10minutes)
Lognormal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE NDJILI (10minutes)
Pearson type 3 (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
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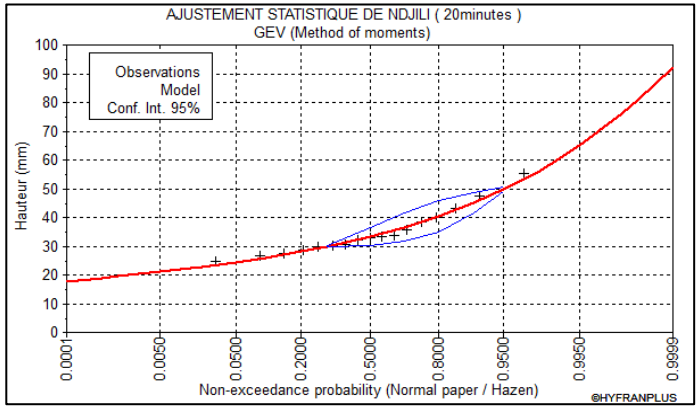
20 MINUTES


AJUSTEMENT STATISTIQUE DE NDJILI (20minutes)
GEV (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
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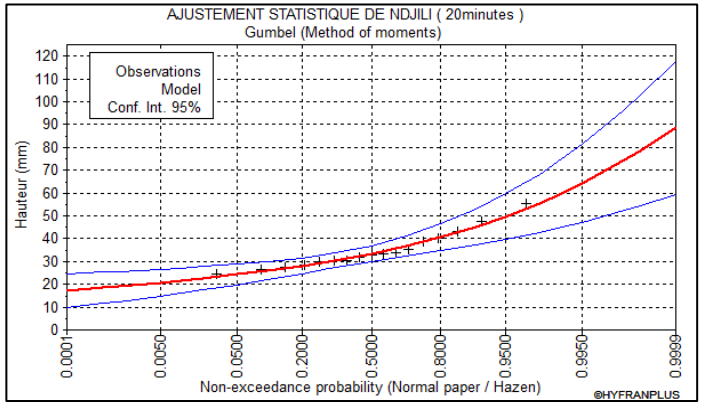
AJUSTEMENT STATISTIQUE DE NDJILI (20minutes)
Gumbel (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
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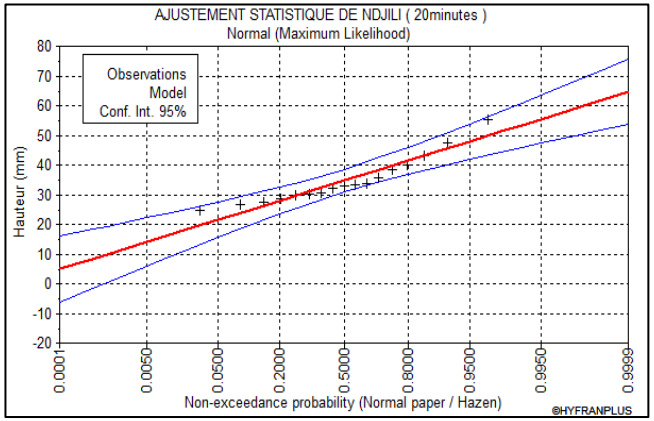
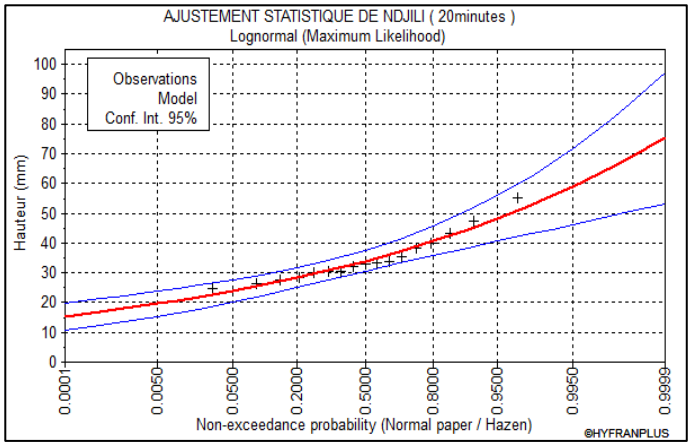
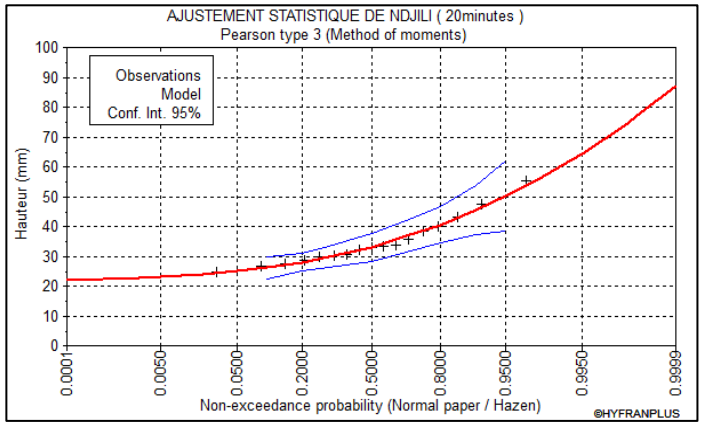
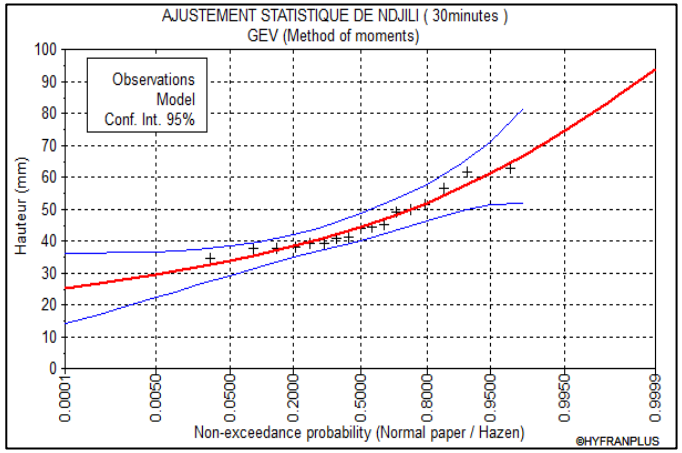

AJUSTEMENT STATISTIQUE DE NDJILI (20minutes)
Normal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE NDJILI (20minutes)
Lognormal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE NDJILI (20minutes)
Pearson type 3 (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

30 MINUTES


AJUSTEMENT STATISTIQUE DE NDJILI (30minutes)
GEV (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
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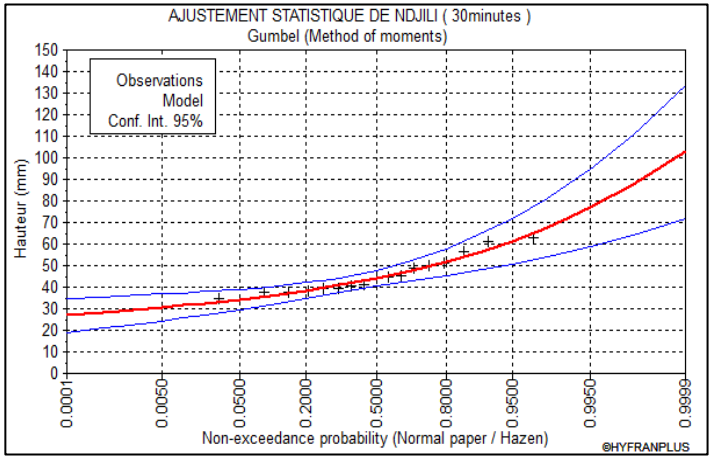
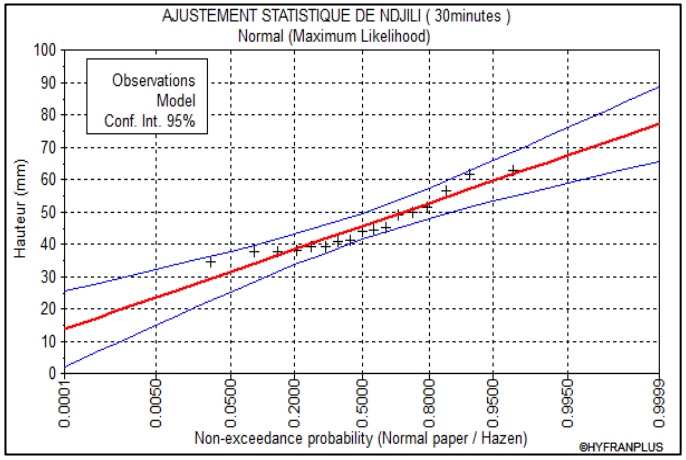
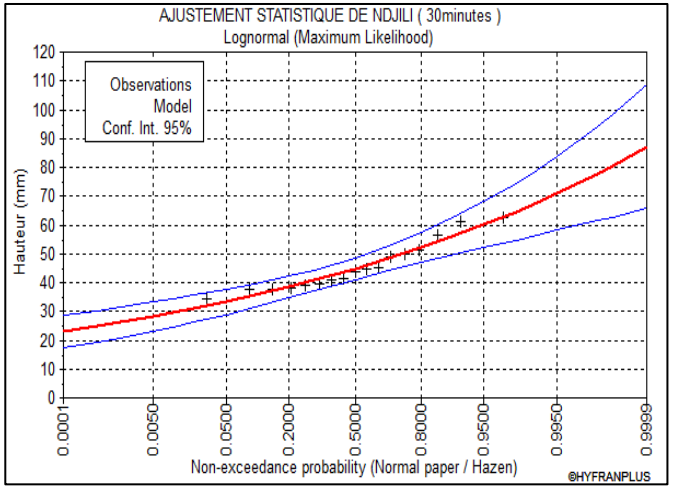
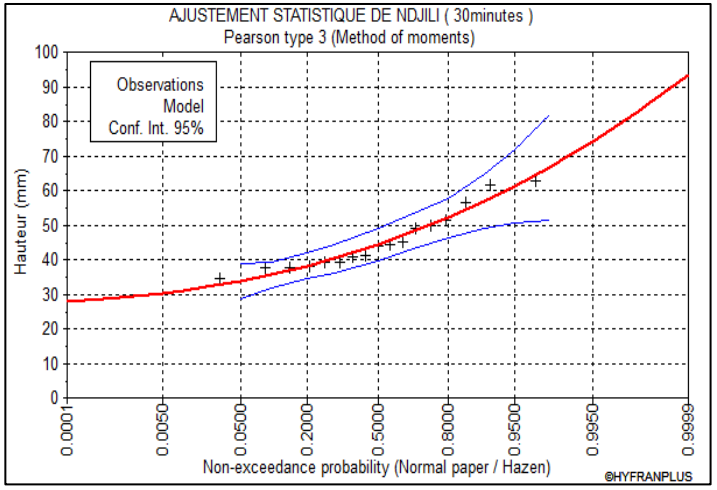
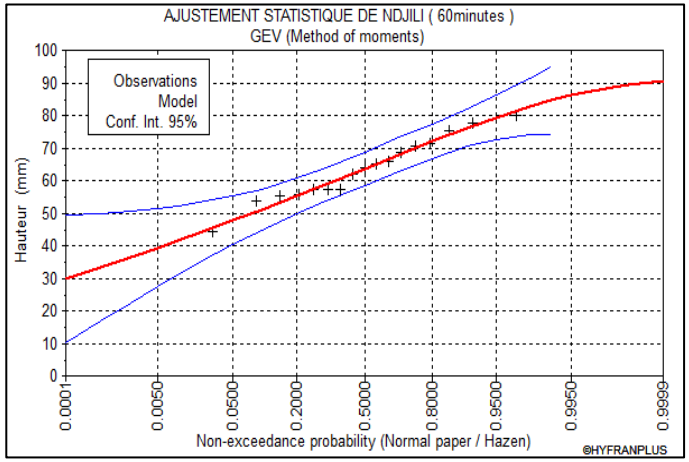

AJUSTEMENT STATISTIQUE DE NDJILI (30minutes)
Gumbel (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

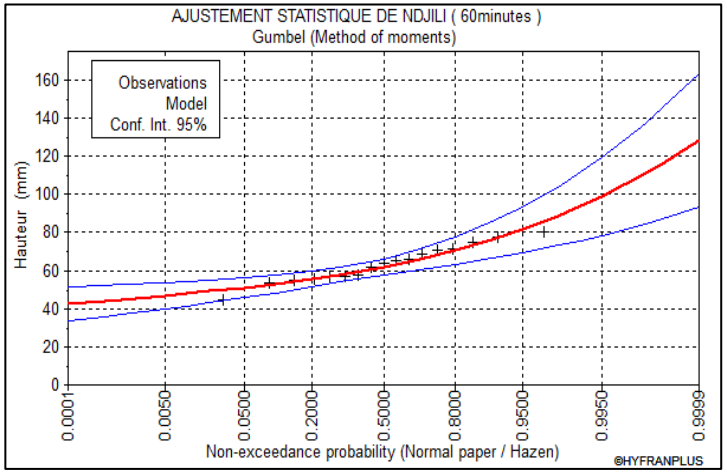

AJUSTEMENT STATISTIQUE DE NDJILI (30minutes)
Normal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

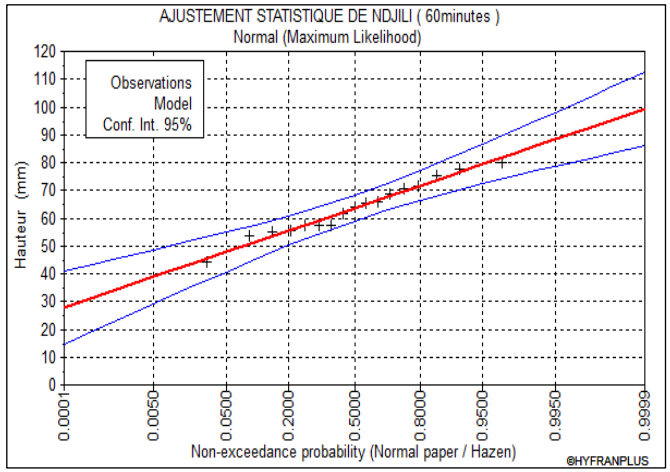

AJUSTEMENT STATISTIQUE DE NDJILI (30minutes)
Lognormal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

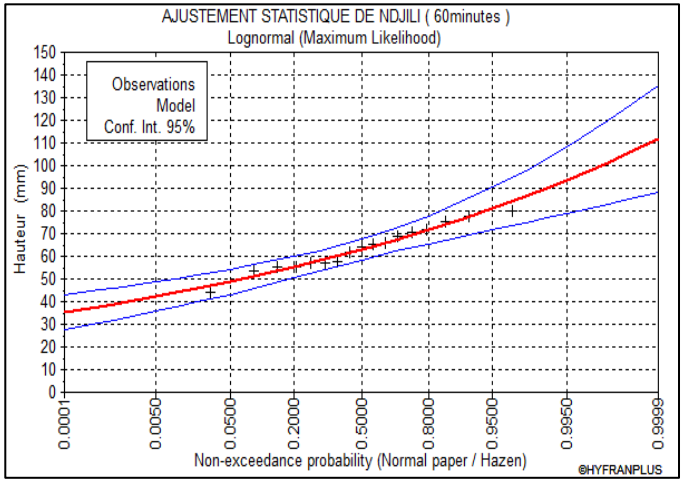

AJUSTEMENT STATISTIQUE DE NDJILI (30minutes)
Pearson type 3 (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

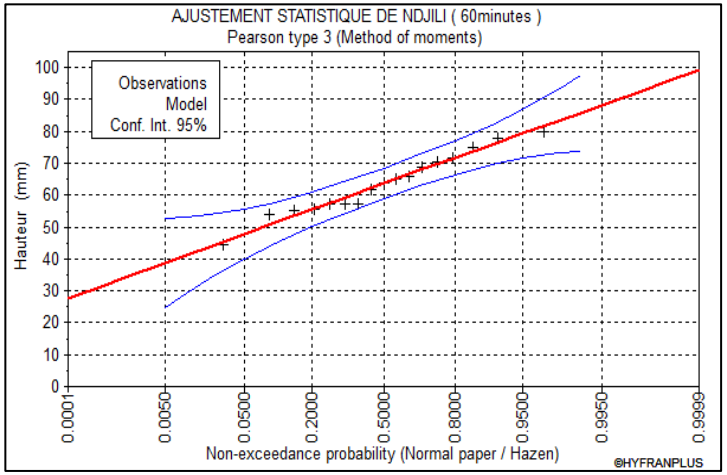
60 MINUTES


AJUSTEMENT STATISTIQUE DE NDJILI (60minutes)
GEV (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE NDJILI (60minutes)
Gumbel (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE NDJILI (60minutes)
Normal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE NDJILI (60minutes)
Lognormal (Maximum Likelihood)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS


AJUSTEMENT STATISTIQUE DE NDJILI (60minutes)
Pearson type 3 (Method of moments)
Observations
Model
Conf. Int. 95%
Hauteur (mm)
Non-exceedance probability (Normal paper / Hazen)
©HYFRANPLUS

IV. Valeurs du coefficient de ruissellement en fonction des caractéristiques du bassin versant (TSHIULA, 2011)

Nature de la couverture	Valeurs de C							
	Petits bassins-Pente moyenne (%)				Bassins moyens-Pente moyenne (%)			
	< 5	5-10	10-30	>30	< 5	5-10	10-30	>30
Plates-formes et chaussées des routes et cours	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95
Terrains dénudés, végétation non couvrante, Terrains érodés, Labours frais	0,80	0,85	0,90	0,95	0,70	0,75	0,80	0,85
Cultures couvrantes, céréales hautes, terrains de parcours, chiendents ras, petite brousse clairsemée	0,75	0,80	0,85	0,90	0,52	0,60	0,72	0,80
Prairies, Brousse dense, Savane à sous-bois	0,70	0,75	0,80	0,85	0,30	0,36	0,42	0,50
Forêt ordinaire en futaie, Sous-bois touffus	0,30	0,50	0,60	0,70	0,13	0,20	0,25	0,30
Grande forêt primaire	0,20	0,25	0,30	0,40	0,15	0,18	0,22	0,25

V. Valeurs des hauteurs extraits des intensités de Kinshasa

t	(5ANS)	(10 ANS)	(20 ANS)	(5O ANS)	(100 ANS)
5	16.51567305	18.53944049	20.4850032	23.0088578	24.90369877
10	25.30222639	28.14532431	30.8718166	34.40085766	37.04561083
15	32.47359197	35.93069305	39.2427383	43.52583717	46.73335306
20	38.76334065	42.72832726	46.5252093	51.43319229	55.10736757
25	44.46951458	48.87475083	53.0923743	58.5431405	62.62295251
30	49.7499662	54.54754738	59.1405628	65.07607383	69.51841277
35	54.70080998	59.85454265	64.7887139	71.1651819	75.9375384
40	59.38594317	64.86725575	70.1155726	76.89846851	81.97521629
45	63.85051172	69.63612001	75.1765811	82.33778999	87.69806882
50	68.127876	74.1983402	80.0125841	87.52864923	93.15505898
55	72.24355194	78.58233039	84.654738	92.50571105	98.3834567
60	76.21759671	82.81039613	89.1274749	97.29612711	103.4124324
65	80.06613121	86.90044376	93.4503893	101.9216499	108.2653401
70	83.80235392	108.7839761	151.737316	230.6407973	372.1924089
75	87.4372402	94.72258187	101.70807	110.7460102	117.515899
80	90.98003903	98.47707924	105.667306	114.9719509	121.9426074
85	94.43863473	113.4811659	146.226107	204.8685978	304.2964714
90	97.81981625	105.7168463	113.294472	123.1043548	130.4556628
95	101.1294814	109.2161356	116.977681	127.0277025	134.5600858
100	104.3727941	112.6429004	120.582545	130.8652793	138.5732334
105	107.5543079	116.0021631	124.11453	134.6231154	142.5015547
110	110.6780633	119.2983777	127.578479	138.3065525	146.3507603
115	113.7476662	122.5355165	130.97871	141.9203495	150.1259362
120	116.7663517	125.7171411	134.319094	145.468769	153.8316361
125	119.7370353	128.8464595	137.603113	148.955647	157.4719569
130	122.6623567	131.9263747	140.833919	152.3844512	161.0506011
135	125.5447149	134.9595234	144.014373	155.7583289	164.5709278
140	128.3862986	137.9483101	147.147083	159.0801472	168.0359966
145	131.189111	140.8949345	150.234434	162.3525273	171.4486035
150	**133.9549917**	**143.8014156**	**153.2786138**	**165.577873**	**174.8113122**

VI. valeurs des débits de pointe (Grady V.2023)

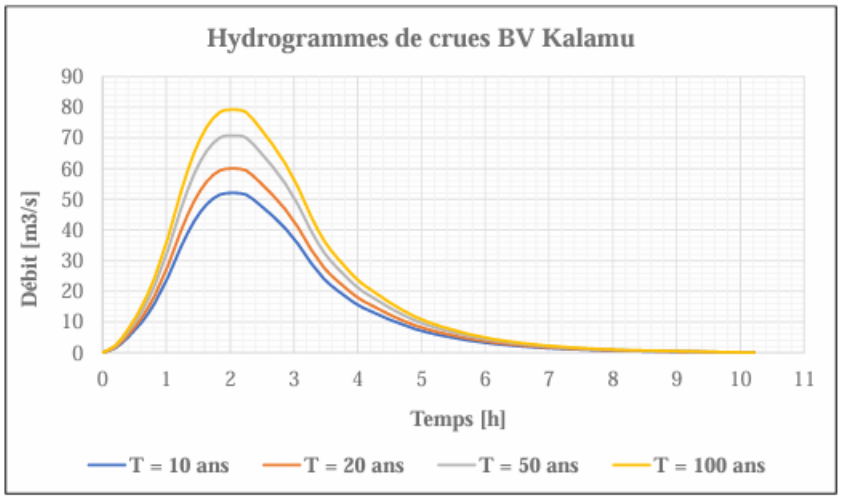


Figure IV-18 : Hydrogrammes de crues du bassin versant de Kalamu

IV.2.4.4 Débits de pointe

Dans un souci de résumer les résultats obtenus sur les débits de pointe des différents hydrogrammes, le tableau suivant nous offre un bref récapitulatif des chiffres.

Tableau IV-15 : Débits de pointe calculés par la méthode de HUS SCS

Cours d'eau		Débits de pointe (m^3/s)			
Rivière	Exutoire	T = 10 ans	T = 20 ans	T = 50 ans	T = 100 ans
Kalamu	Pont Lumumba	52.04	59.99	70.78	79.18

vii calcul de Kolmogorov-Smirnov (BINZA)

		N	Hmax		Fn		Z	EXP	Fx	D+	D-	
		1	9.0	9	0.03125	0	1.3	3.5005437	0.030181	0.001069	0.03	
		2	10.0	10	0.0625	0.03125	0.8638132	2.3721892	0.0932763	-0.0307763	0.06	
ϵ =	2.57	3	10.0	10	0.09375	0.0625	0.8638132	2.3721892	0.0932763	0.0004737	0.0307763	
ϋ=,	12.22	4	10.0	10	0.125	0.09375	0.9	2.3721892	0.0932763	0.0317237	-0.0004737	
		5	10.0	10	0.15625	0.125	0.9	2.3721892	0.0932763	0.0629737	-0.0317237	
		6	10.1	10.1	0.1875	0.15625	0.8	2.2816588	0.1021147	0.0853853	-0.0541353	
		7	10.2	10.2	0.21875	0.1875	0.7859922	2.1945834	0.111405	0.107345	-0.076095	
		8	10.6	10.6	0.25	0.21875	0.6303502	1.8782682	0.1528546	0.0971454	-0.0658954	
		9	10.8	10.8	0.28125	0.25	0.5525292	1.7376423	0.1759347	0.1053153	-0.0740653	
		10	11.3	11.3	0.3125	0.28125	0.3579767	1.4304322	0.2392055	0.0732945	-0.0420445	
		11	11.5	11.5	0.34375	0.3125	0.2801556	1.3233358	0.2662457	0.0775043	-0.0462543	
		12	12.0	12	0.375	0.34375	0.0856031	1.0893739	0.3364271	0.0385729	-0.0073229	
		13	12.0	12	0.40625	0.375	0.0856031	1.0893739	0.3364271	0.0698229	-0.0385729	
		14	12.5	12.5	0.4375	0.40625	-0.1089494	0.8967758	0.4078826	0.0296174	0.0016326	
		15	12.7	12.7	0.46875	0.4375	-0.1867704	0.8296342	0.4362088	0.0325412	-0.0012912	
		16	12.8	12.8	0.5	0.46875	-0.2256809	0.7979727	0.4502408	0.0497592	-0.0185092	
		17	12.9	12.9	0.53125	0.5	-0.2645914	0.7675195	0.464163	0.067087	-0.035837	
		18	13.6	13.6	0.5625	0.53125	-0.536965	0.5845196	0.5573736	0.0051264	0.0261236	
		19	14.0	14	0.59375	0.5625	-0.692607	0.5002702	0.6063668	-0.0126168	0.0438668	
		20	14.2	14.2	0.625	0.59375	-0.770428	0.4628149	0.6295091	-0.0045091	0.0357591	
		21	14.5	14.5	0.65625	0.625	-0.8871595	0.4118239	0.6624409	-0.0061909	0.0374409	
		22	14.5	14.5	0.6875	0.65625	-0.8871595	0.4118239	0.6624409	0.0250591	0.0061909	
		23	15.5	15.5	0.71875	0.6875	-1.2762646	0.2790778	0.756481	-0.037731	0.068981	
		24	16.0	16	0.75	0.71875	-1.4708171	0.2297377	0.794742	-0.044742	0.075992	
		25	17.0	17	0.78125	0.75	-1.8599222	0.1556847	0.855829	-0.074579	0.105829	
		26	17.2	17.2	0.8125	0.78125	-1.9377432	0.1440286	0.865863	-0.053363	0.084613	
		27	17.5	17.5	0.84375	0.8125	-2.0544747	0.1281601	0.8797125	-0.0359625	0.0672125	
		28	17.8	17.8	0.875	0.84375	-2.1712062	0.11404	0.8922223	-0.0172223	0.0484723	
		29	18.6	18.6	0.90625	0.875	-2.4824903	0.0835349	0.9198589	-0.0136089	0.0448589	
		30	19.3	19.3	0.9375	0.90625	-2.7548638	0.0636177	0.9383637	-0.0008637	0.0321137	
		31	19.8	19.8	0.96875	0.9375	-2.9494163	0.0523703	0.9489774	0.0197726	0.0114774	
		32	19.9	19.9	1	0.96875	-2.9883268	0.0503716	0.950876	0.049124	-0.017874	
										D+(max)	D-(max)	
										0.107345	0.105829	
									Dn MAX=	0.107345		
									Dcric=	0.281	0.01	OK
										0.234	0.05	OK

vii. 2 calcul de Kolmogorov-Smirnov (NDJILI)

		N	H max	Fn		Z	EXP	Fx	D+	D-	
		1	9.9	0.0588235	0	0.7	2.0333352	0.1308982	-0.0720747	0.1308982	
		2	10.0	0.1176471	0.0588235	0.7	1.9687904	0.1396256	-0.0219786	0.0808021	
ϵ =	3.1	3	10.1	0.1764706	0.1176471	0.6	1.9062945	0.1486301	0.0278405	0.0309831	
$\dot{\upsilon}$=	12.1	4	10.1	0.2352941	0.1764706	0.6	1.9062945	0.1486301	0.086664	-0.0278405	
		5	10.4	0.2941176	0.2352941	0.5	1.7304597	0.1772029	0.1169147	-0.0580912	
		6	10.8	0.3529412	0.2941176	0.4	1.52098	0.2184977	0.1344435	-0.07562	
		7	10.9	0.4117647	0.3529412	0.4	1.472699	0.2293058	0.182459	-0.1236354	
		8	11.5	0.4705882	0.4117647	0.2	1.2135481	0.2971411	0.1734471	-0.1146236	
		9	14.0	0.5294118	0.4705882	-0.6	0.5417757	0.5817144	-0.0523026	0.1111262	
		10	14.0	0.5882353	0.5294118	-0.6	0.5417757	0.5817144	0.0065209	0.0523026	
		11	15.0	0.6470588	0.5882353	-0.9	0.392396	0.6754366	-0.0283778	0.0872013	
		12	15.0	0.7058824	0.6470588	-0.9	0.392396	0.6754366	0.0304457	0.0283778	
		13	15.5	0.7647059	0.7058824	-1.1	0.3339466	0.716092	0.0486139	0.0102097	
		14	19.6	0.8235294	0.7647059	-2.4	0.088979	0.9148648	-0.0913354	0.1501589	
		15	20.0	0.8823529	0.8235294	-2.5	0.0782077	0.9247723	-0.0424194	0.1012429	
		16	20.0	0.9411765	0.8823529	-2.5	0.0782077	0.9247723	0.0164041	0.0424194	
		17	20.6	1.0000	0.9411765	-2.7	0.0644455	0.9375872	0.0624128	-0.0035893	
									D+max	D-max	
									0.182459	0.1501589	
								Dn max	0.182459		
								Dcritique	0.381	1%	OK
									0.318	5%	OK